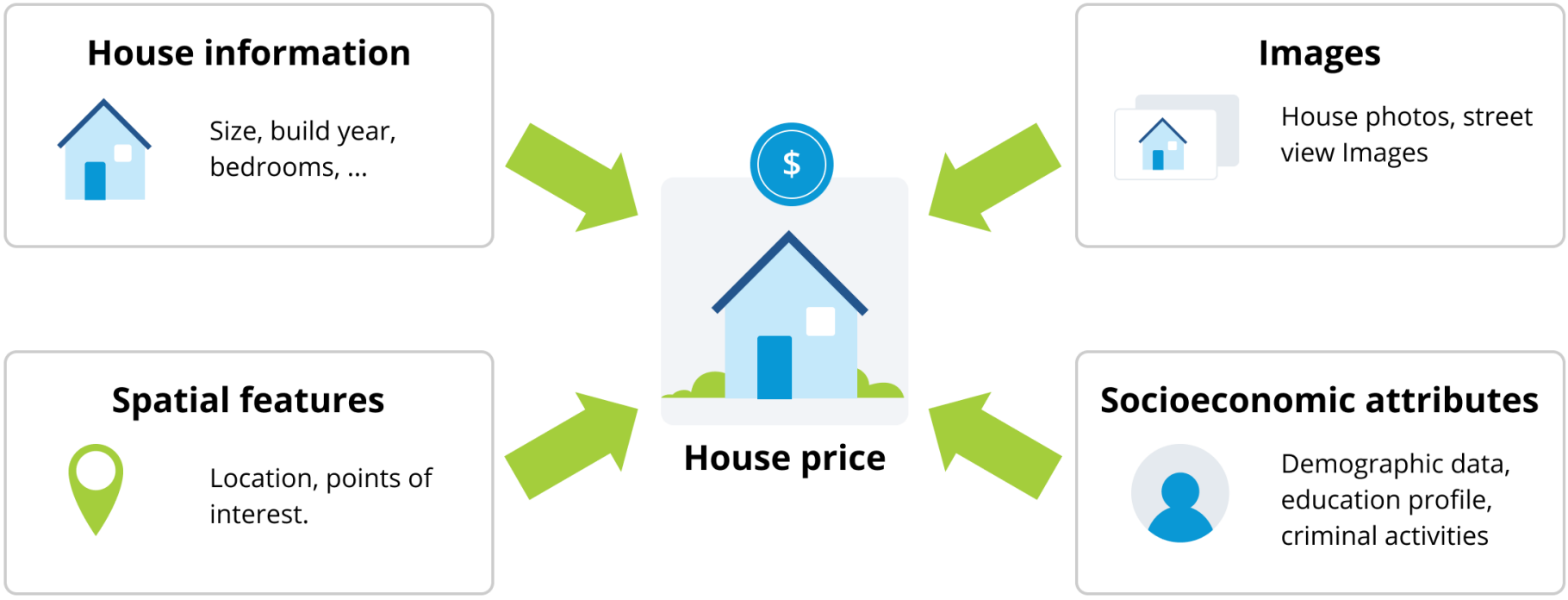
Linear Regression

Introduction

Example of a house price prediction problem.



Introduction

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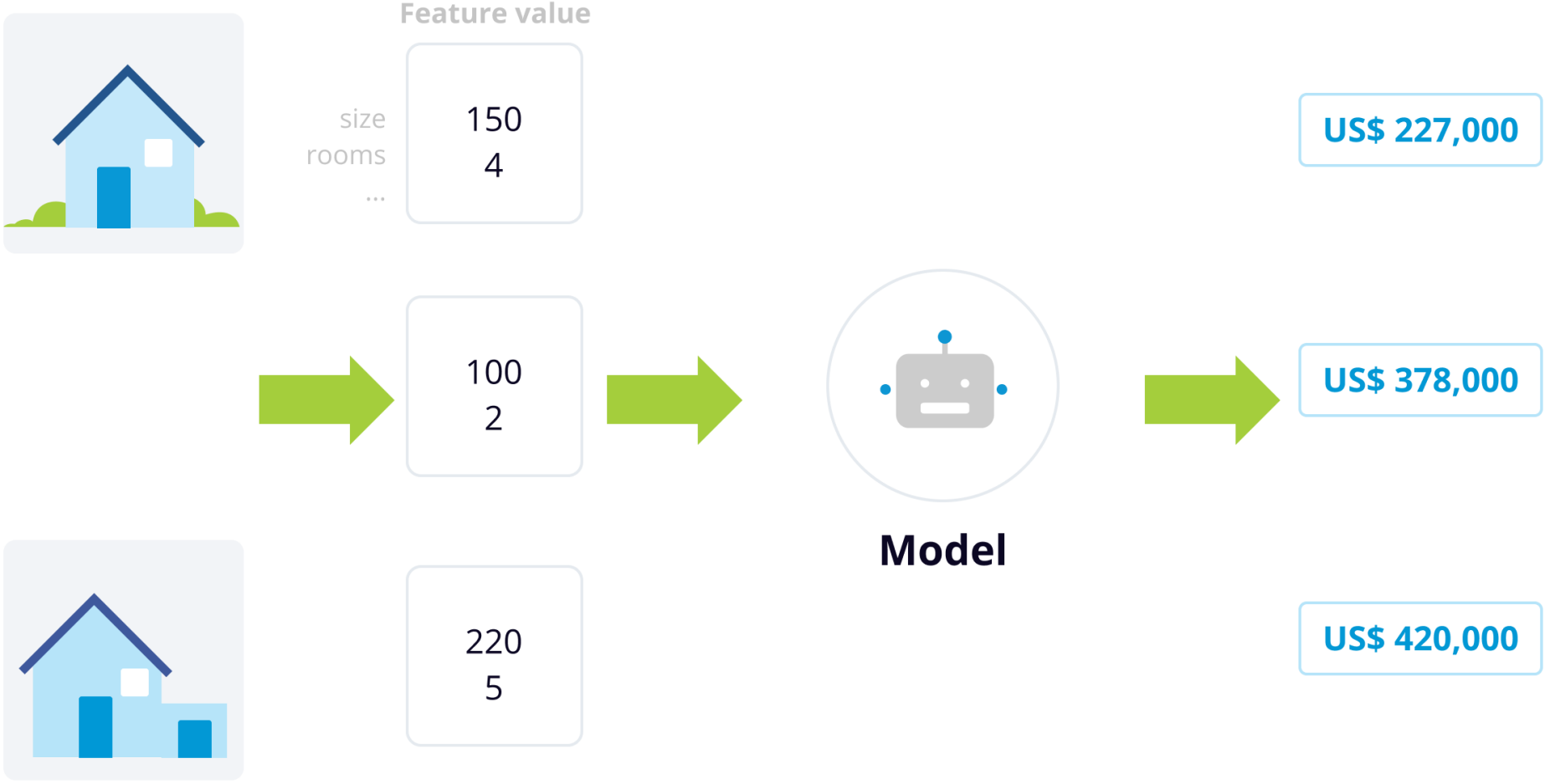
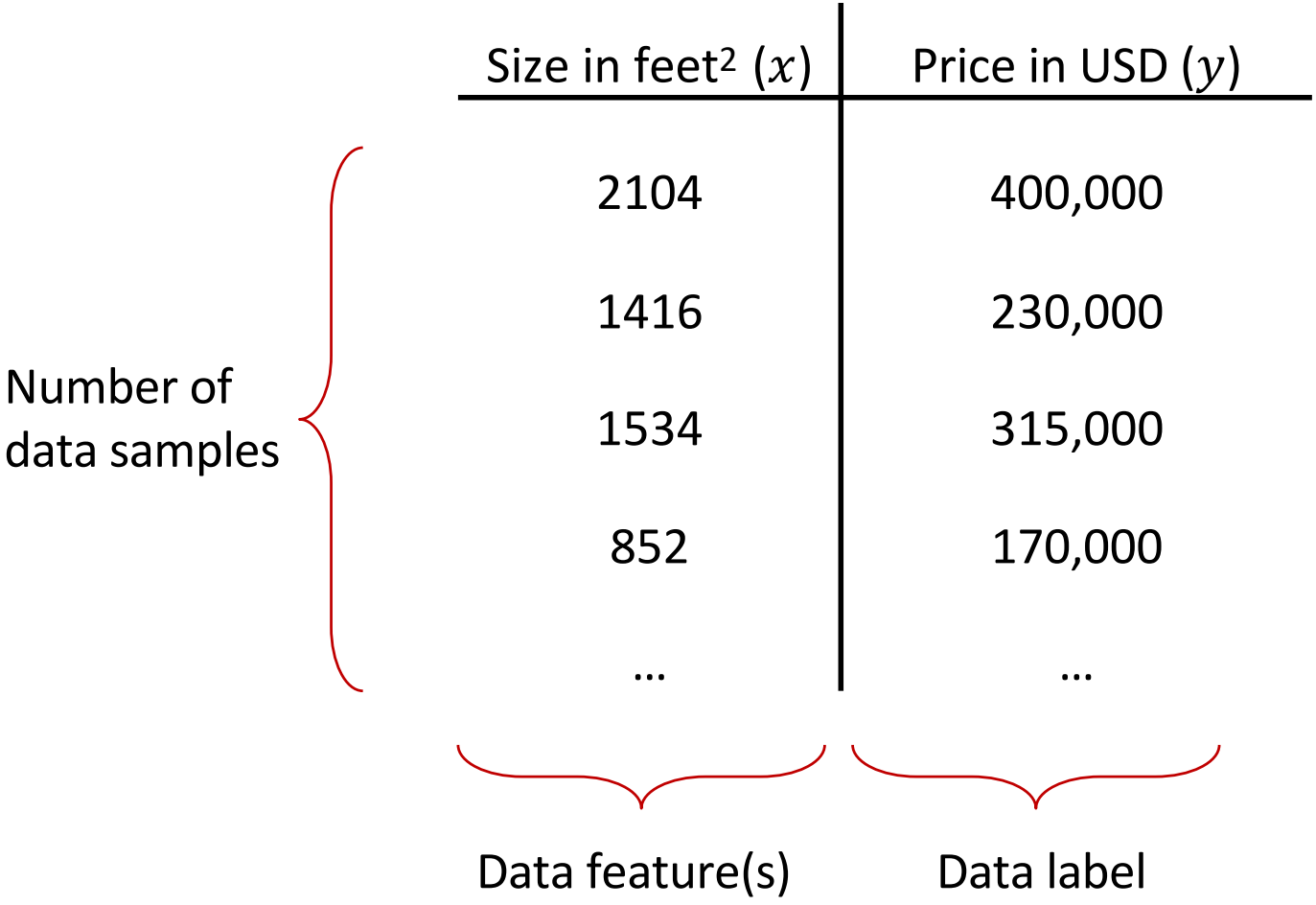


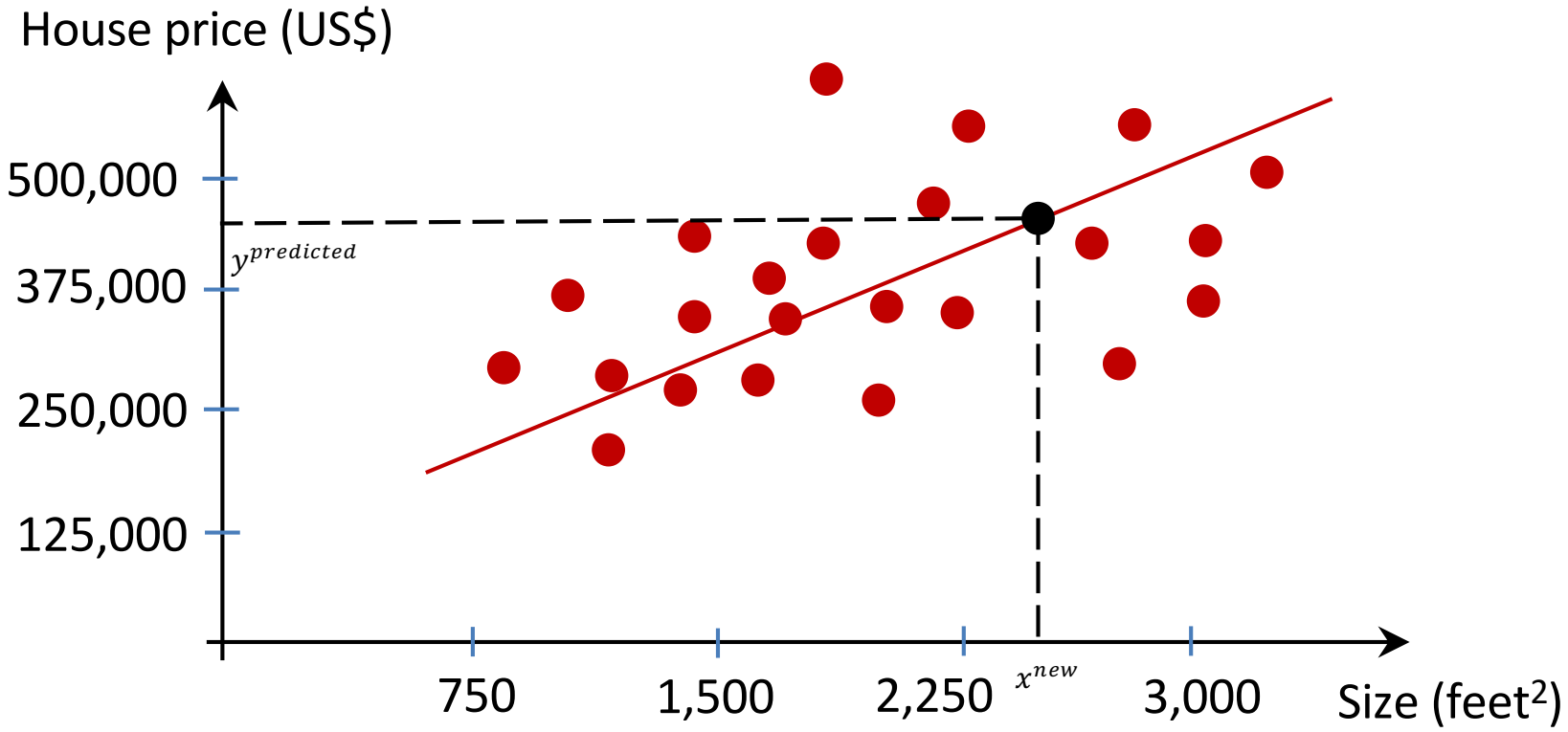
Problem Setup

Example of a house price prediction problem.



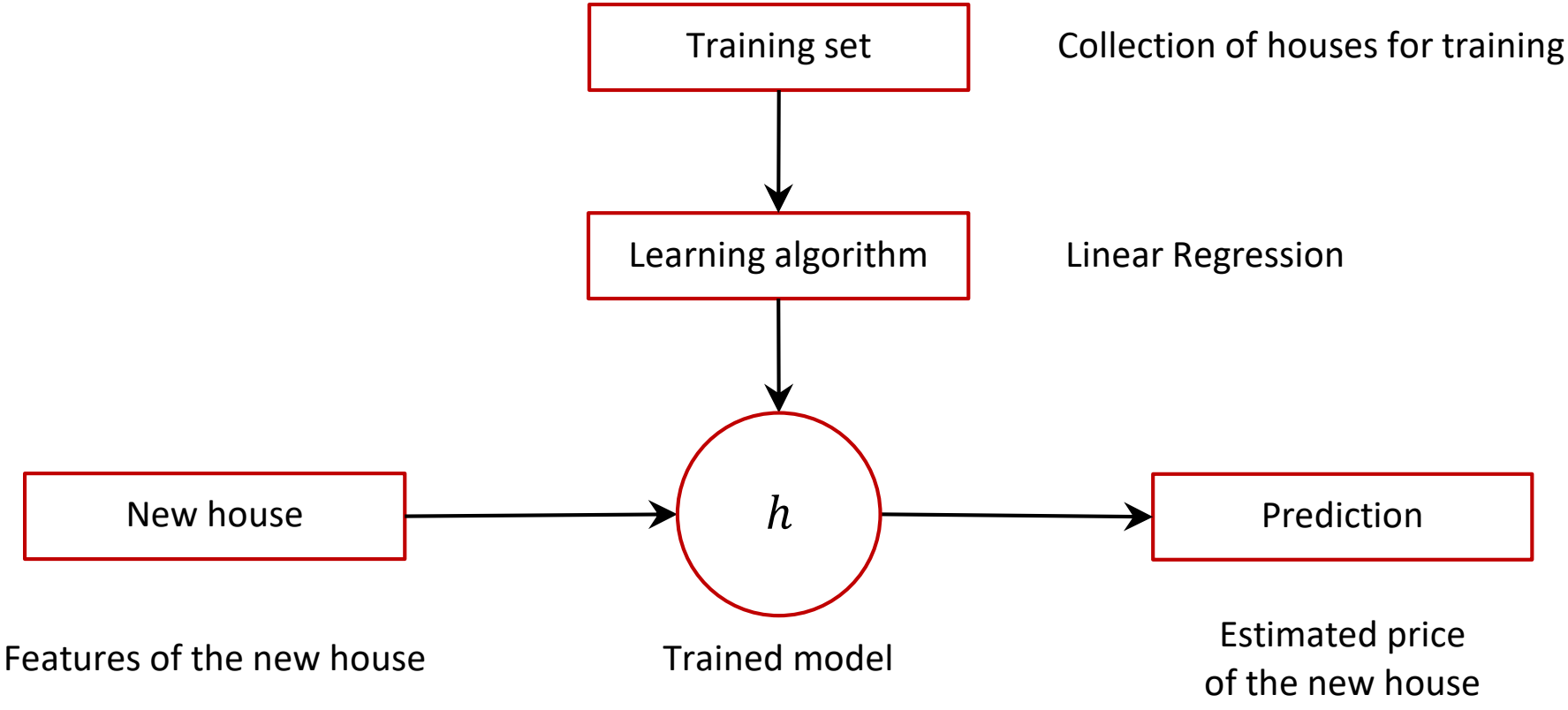
Problem Setup

Objective: Find a line, meaning a function, that learns the relationship between the input features and the target labels.



Question: What desired properties should the fitting function possess?

Problem Setup



Problem Setup

Size in feet² (x)	Price in USD (y)
2104	400,000
1416	230,000
1534	315,000
852	170,000
…	…

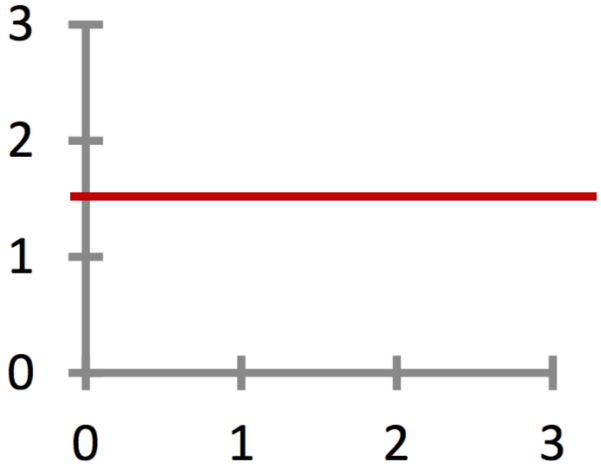
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$
where θ_0, θ_1 are learnable parameters.

Questions:

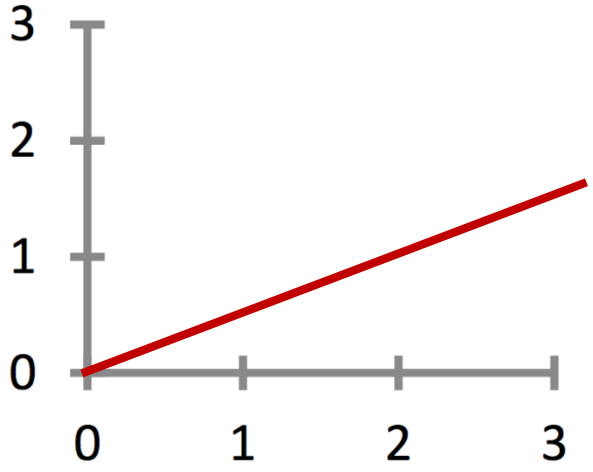
- What is the role of θ_0 and θ_1?
- How to choose good values of θ_0 and θ_1?

Observation

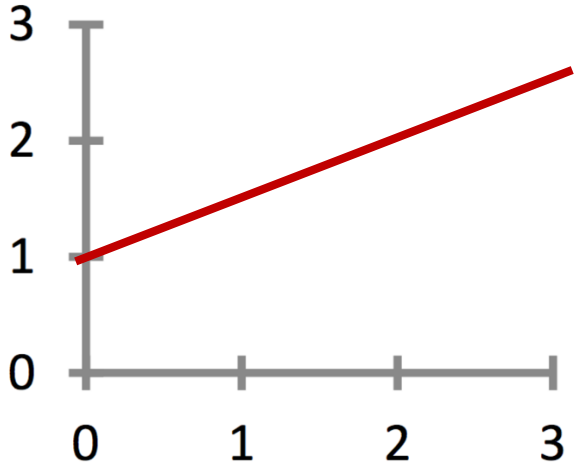
Fitting function $h_\theta(x) = \theta_0 + \theta_1 x$



$\theta_0 = 1.5$
$\theta_1 = 0$



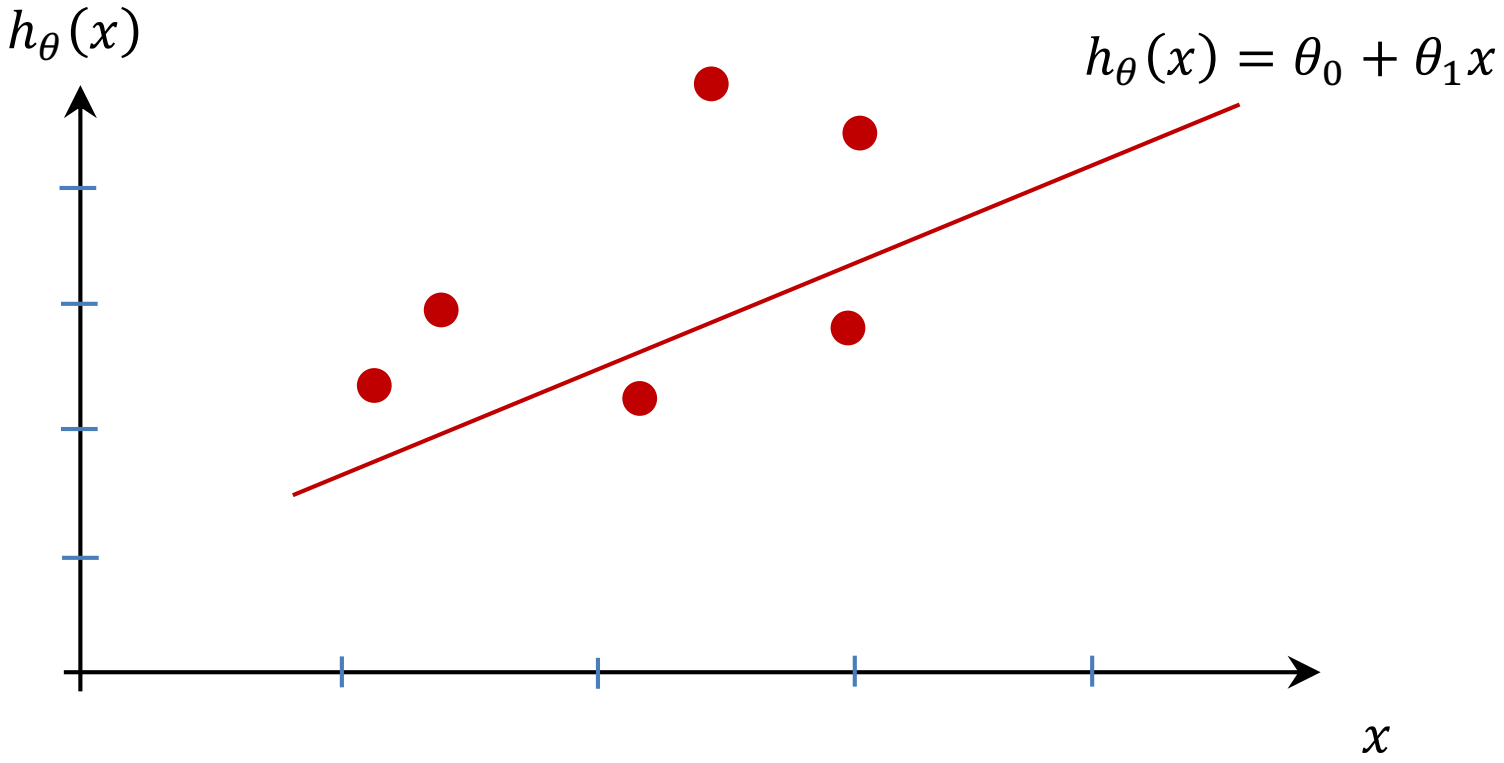
$\theta_0 = 0$
$\theta_1 = 0.5$



$\theta_0 = 1$
$\theta_1 = 0.5$

Visualization of the fitting function with some random values of θ_0 and θ_1

Observation

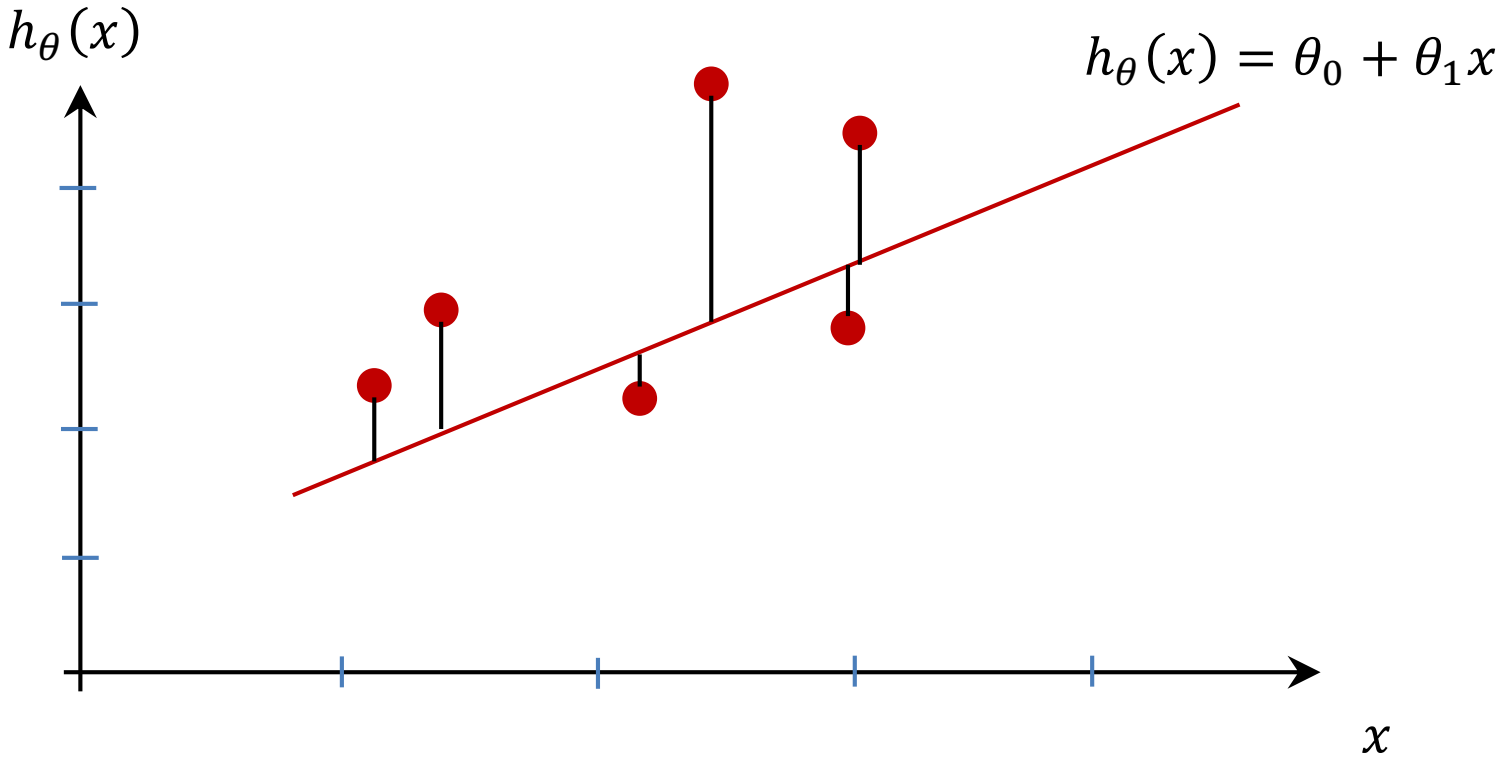


Question: How to choose good values of θ_0 and θ_1?

Answer: Good values of θ_0 and θ_1 make $h_\theta(x)$ *close* to y for training samples (x, y)

Follow-up question: what does *close* mean?

Observation



Follow-up question: What does *close* mean?

Answer: We measure closeness using the mean squared error, or MSE. It tells us how well the model fits the data. MSE is the average of all squared errors between the predicted values and the true values.

Linear Regression

General form of the hypothesis function:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_d x_d = \sum_{j=0}^{d} \theta_j x_j$$

where input data feature is d-dimensional, i.e., x_1, …, x_d.

We train the model by minimizing the MSE, written as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left(h_\theta\left(x^{(i)}\right) - y^{(i)} \right)^2$$

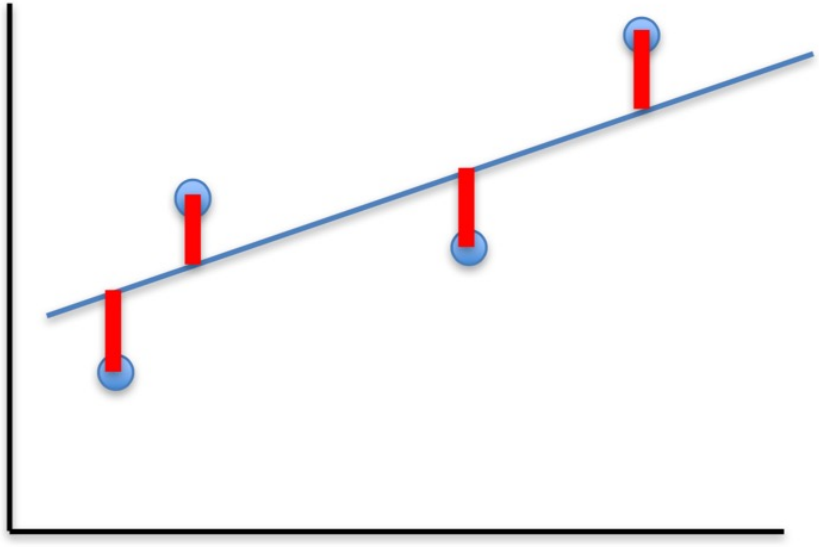
A smaller MSE means the model performs better.

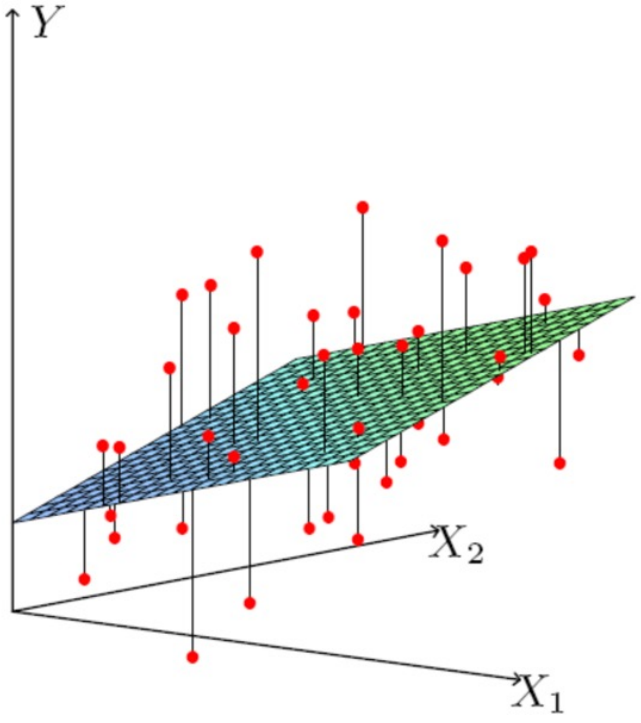
Least Squares Linear Regression

Loss function:

$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

Fit by solving $\min_\theta J(\theta)$





Intuition Behind Loss Function

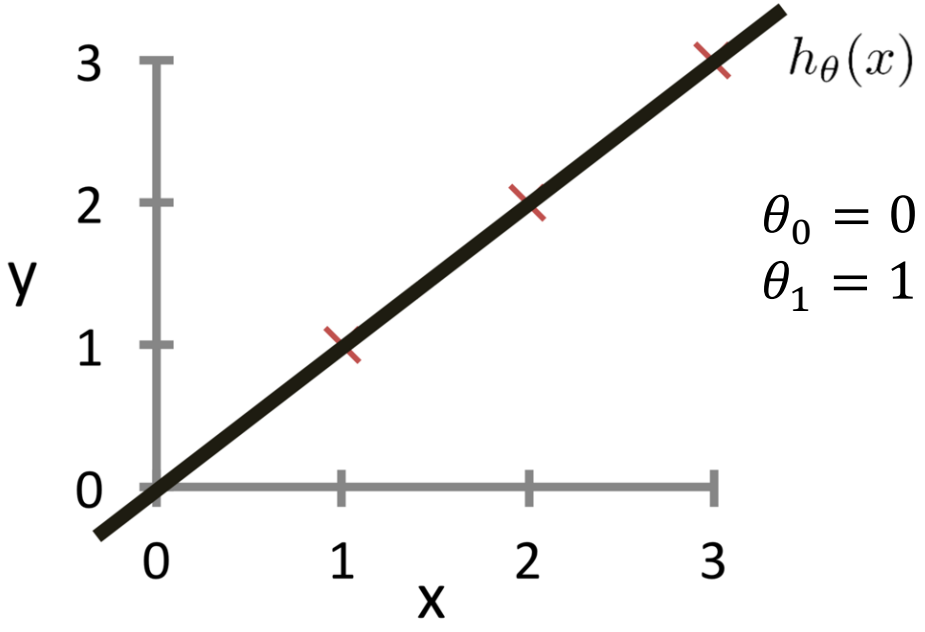
Loss function:

$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

For insight on $J(\theta)$, let's assume that $\theta = [\theta_0, \theta_1]$

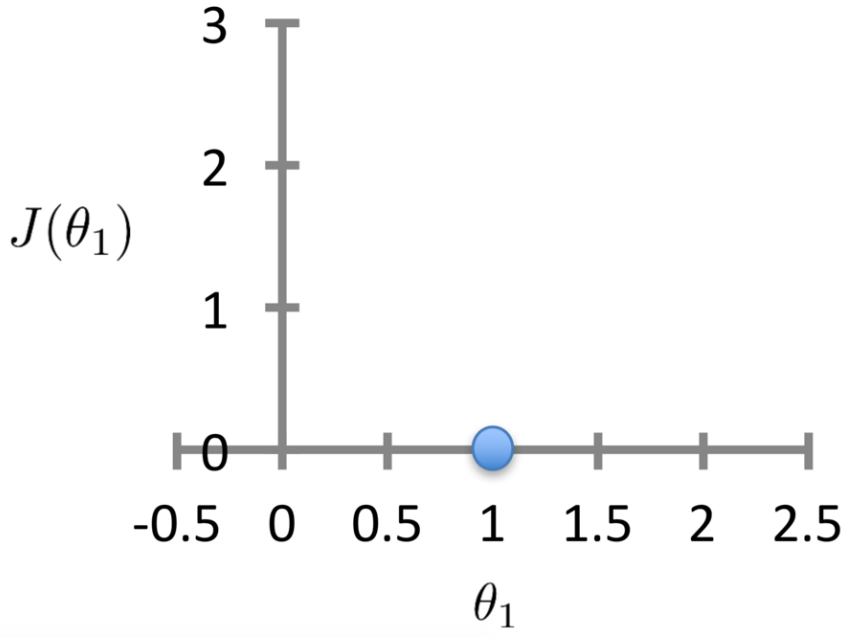
$h_\theta(x)$

(for fixed θ_1, this is a function of x)



$J(\theta_1)$

(function of the parameter θ_1)



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$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

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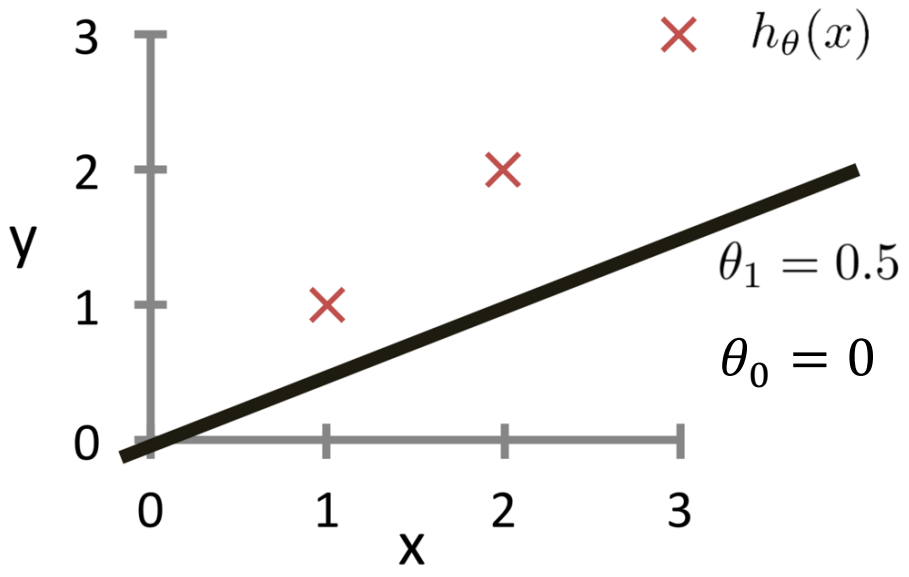

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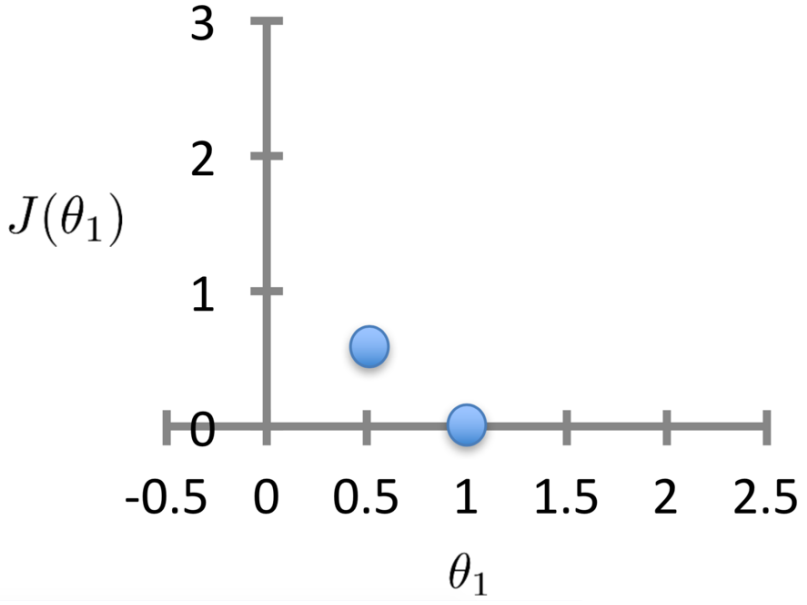

$$J[0,0.5] = \frac{1}{2\times 3}\left[(0.5-1)^2+(1-2)^2+(1.5-3)^2\right] \approx 0.58$$

Intuition Behind Loss Function

Loss function:

$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$h_\theta(x)$

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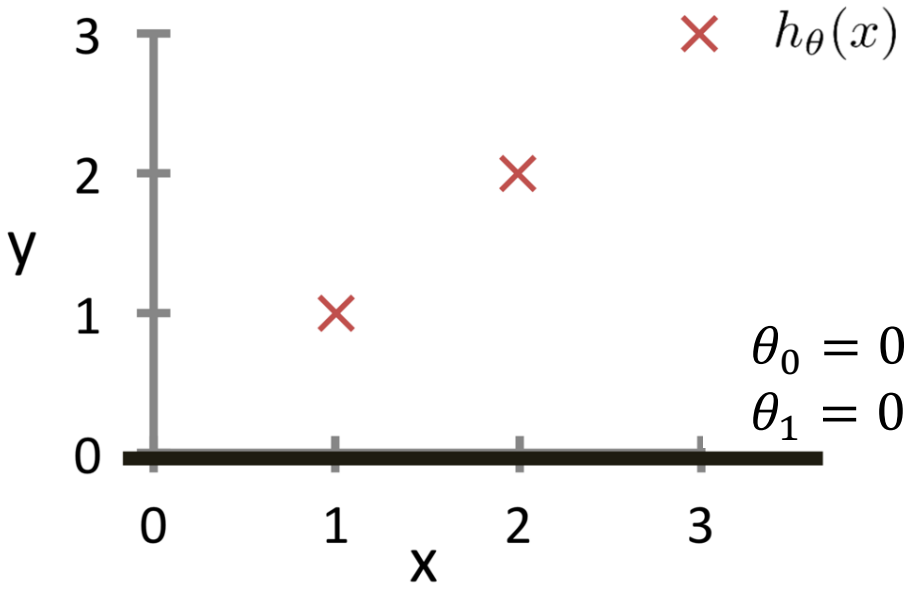

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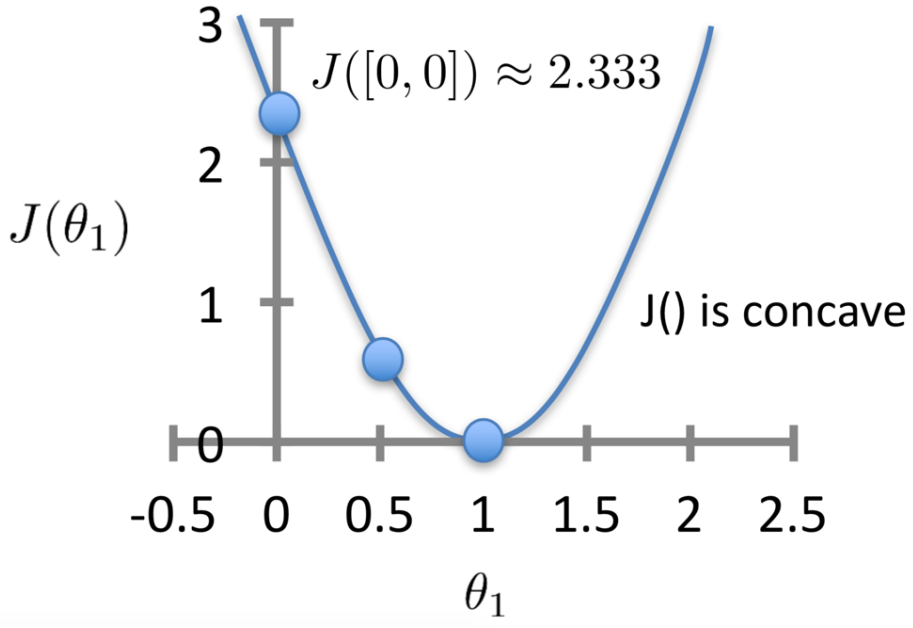
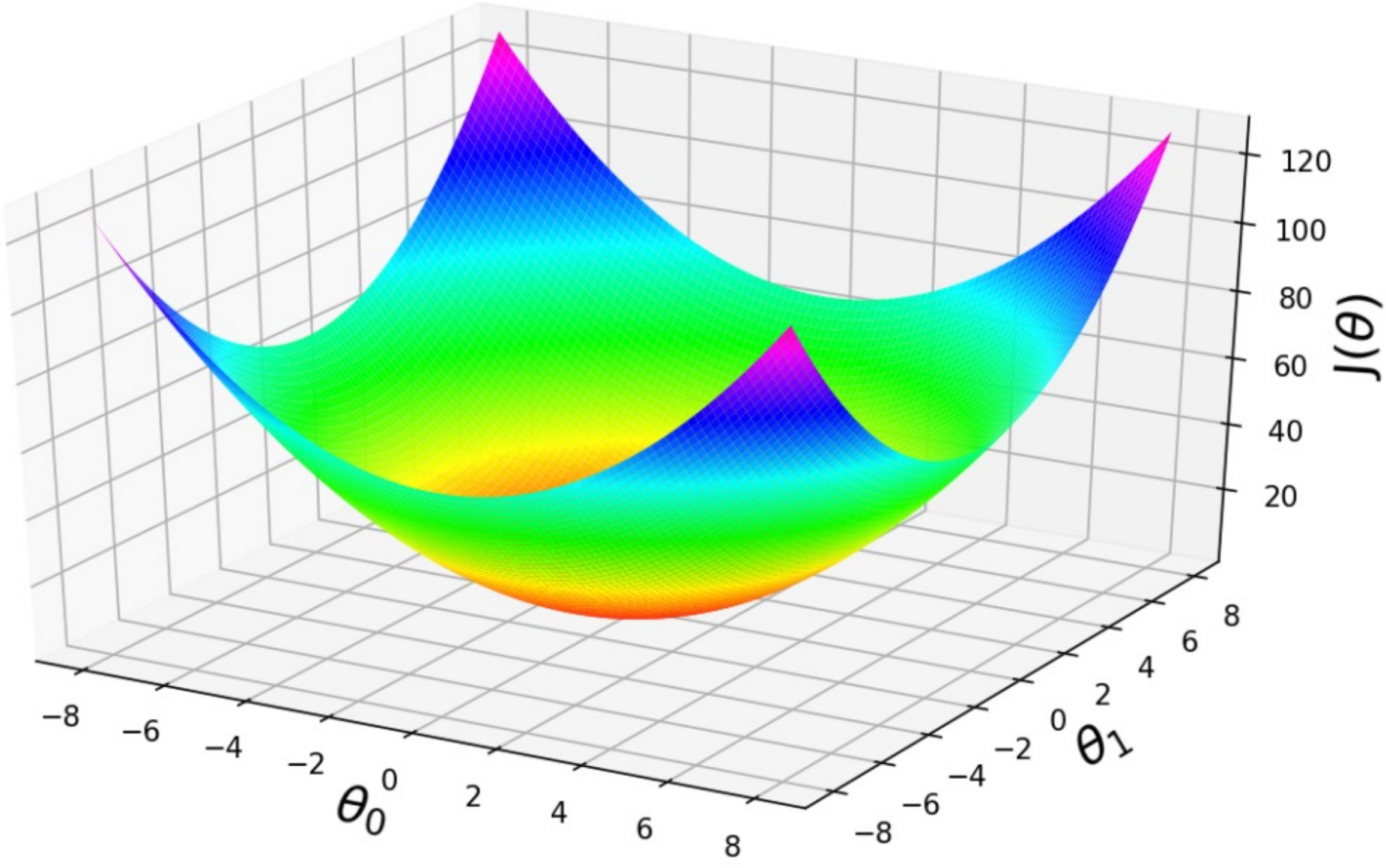


$$J[0,0] = \frac{1}{2\times 3}[(0-1)^2+(0-2)^2+(0-3)^2] \approx 2.333$$

Intuition Behind Loss Function



Formulation

- Hypothesis:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

- Parameters:

$$\theta_0, \theta_1$$

- Loss Function:

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^{n} \left(h_\theta\left(x^{(i)}\right) - y^{(i)} \right)^2$$

- Goal:

$$\underset{\theta_0, \theta_1}{\text{minimize}}\, J(\theta_0, \theta_1)$$

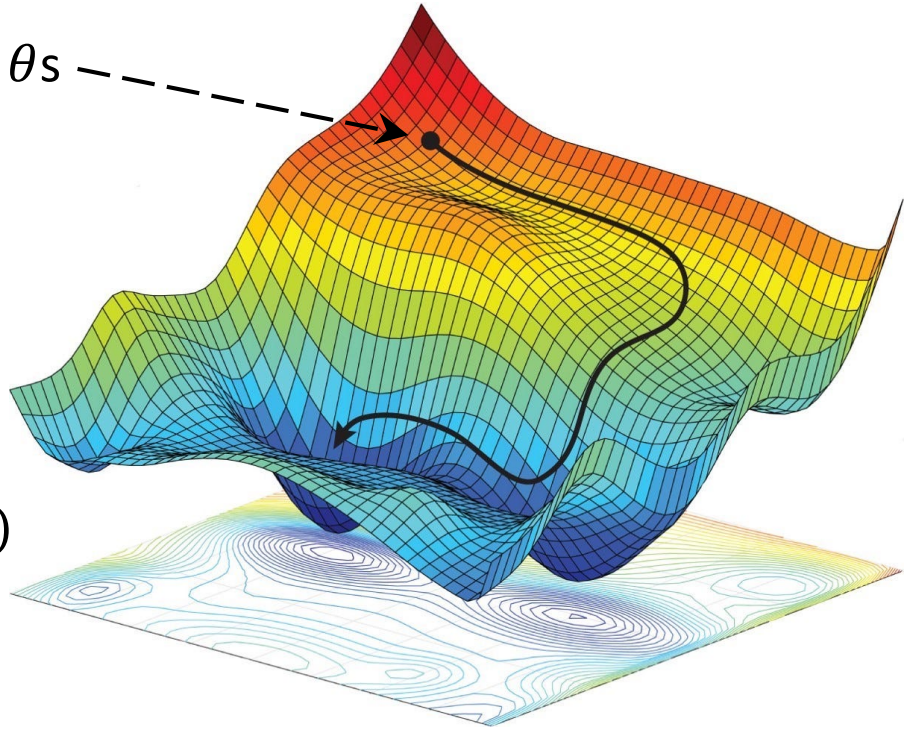
An optimization problem

Have some function $J(\theta_0, \theta_1)$ and we want $\underset{\theta_0,\theta_1}{\text{minimize}}\, J(\theta_0, \theta_1)$

Solution:

- Start with some θ_0, θ_1.
- Keep adjusting θ_0, θ_1 to reduce until we hopefully end up at a minimum $J(\theta_0, \theta_1)$

Question: how to properly adjust θ_0, θ_1 to reduce $J(\theta)$?



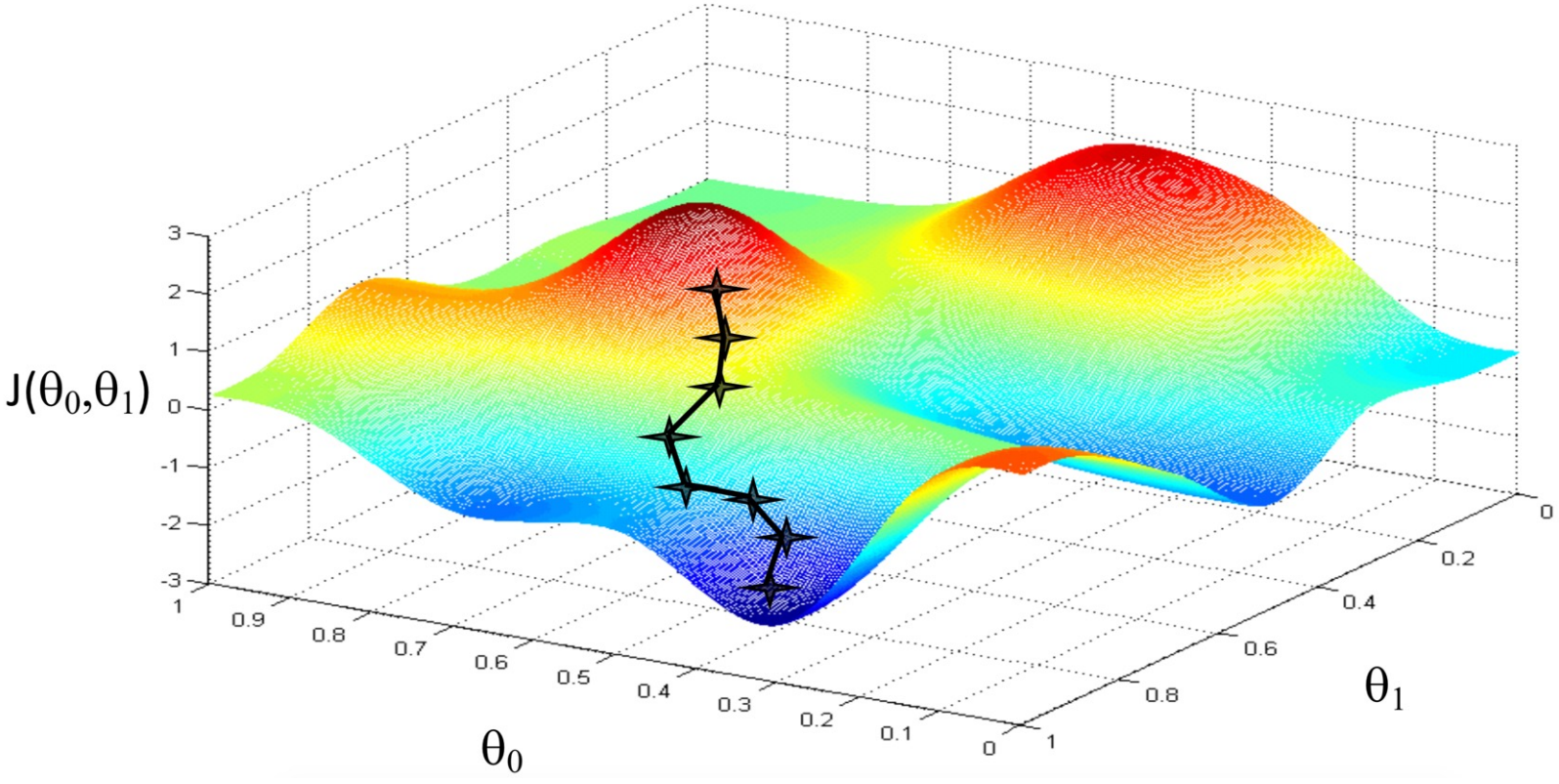
Example visualization of a loss surface $J(\theta_0, \theta_1)$

An optimization problem

Question: how to properly adjust θ_0, θ_1 to reduce $J(\theta)$?

Idea: Starting from an initial value, the optimizer aims to gradually move in the direction that leads a to a better (lower) value of the loss function $J(\theta)$.

→ *Gradient Descent algorithm*



Gradient Descent

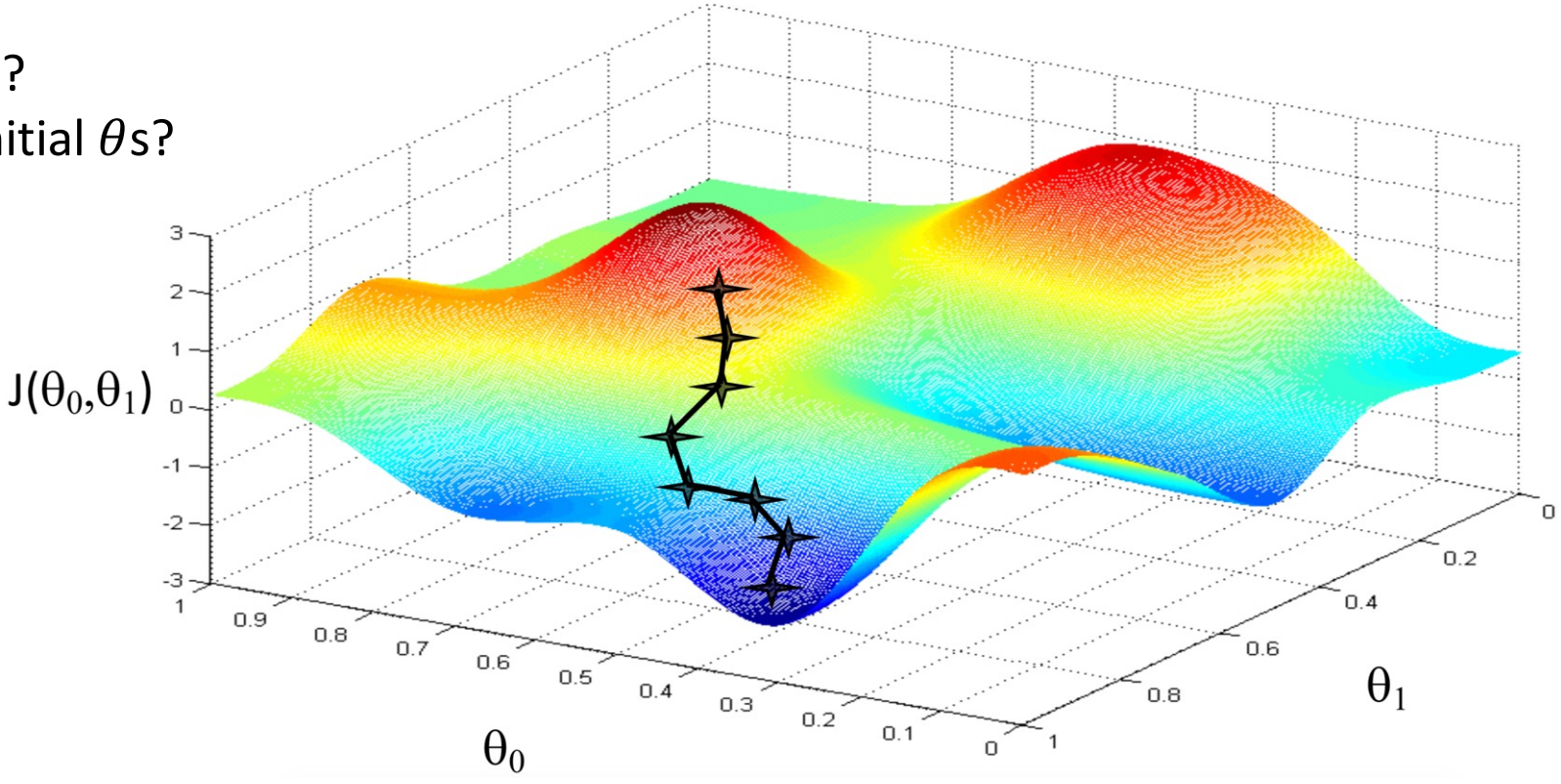
The Gradient Descent algorithm:

- Choose initial values for θs.
- Until we reach the minimum, update θs to reduce $J(\theta)$:

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

Questions:

- What is α?
- Why minus?
- Different initial θs?



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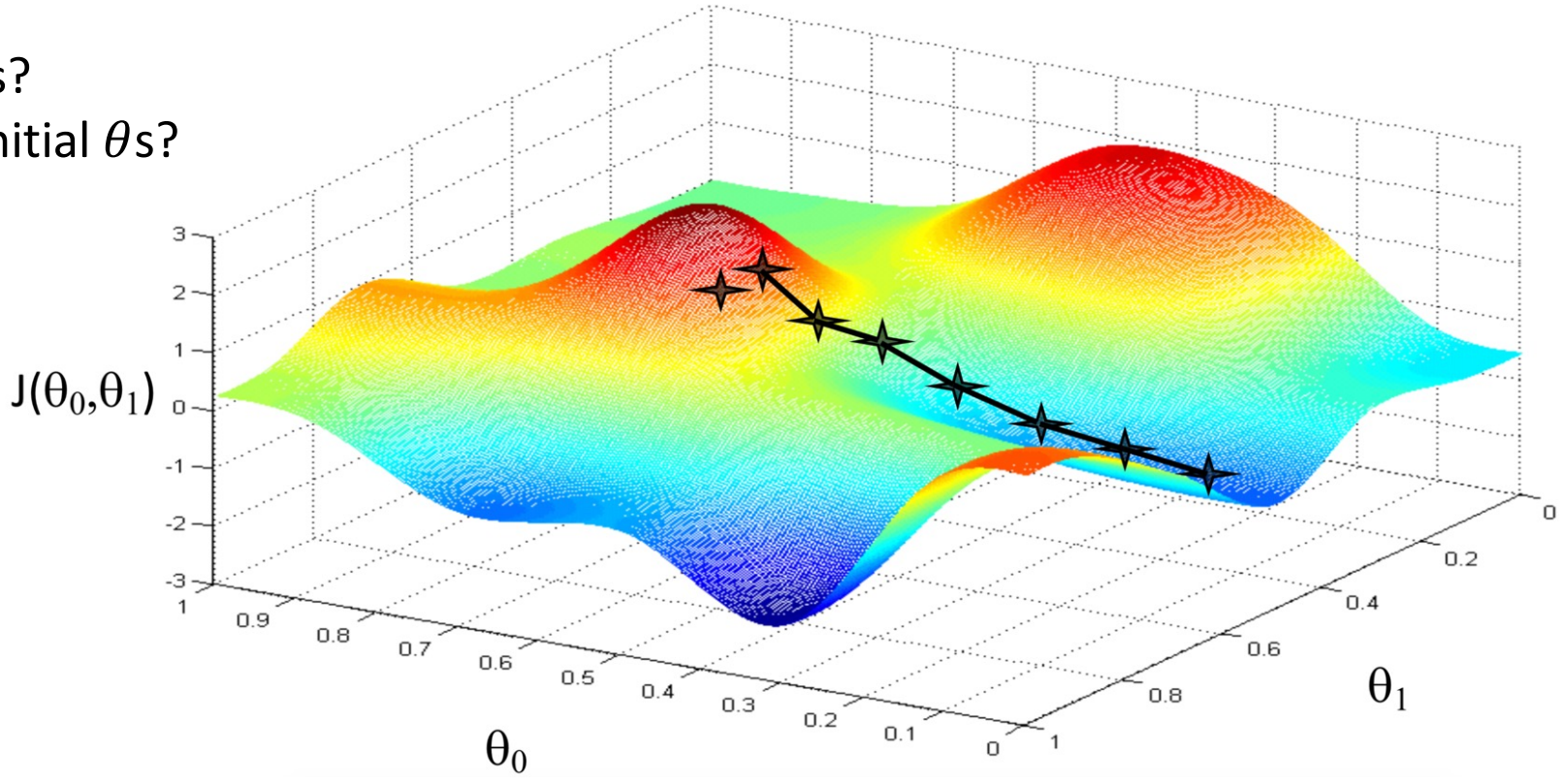

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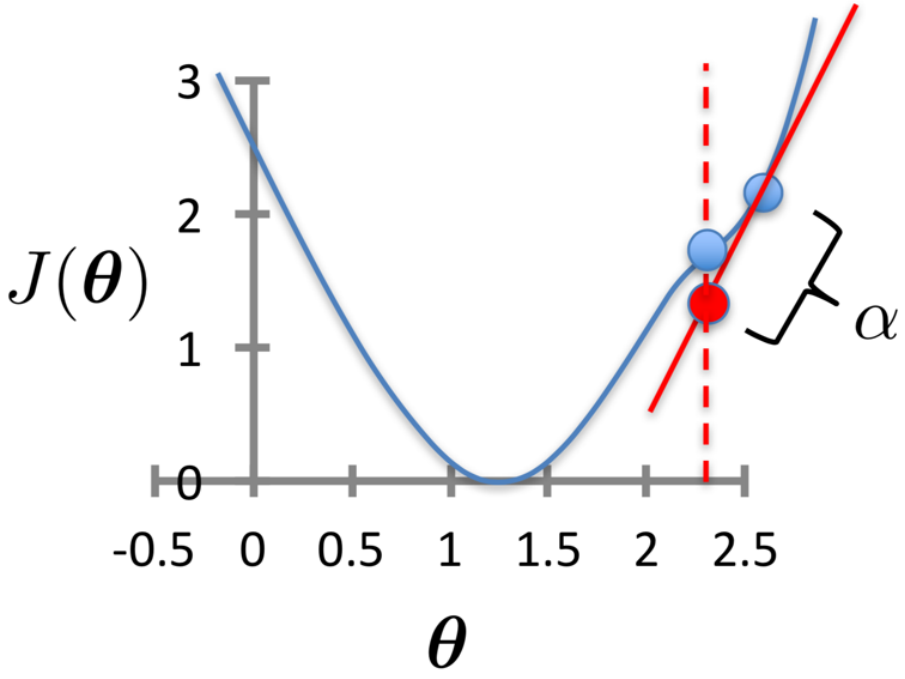

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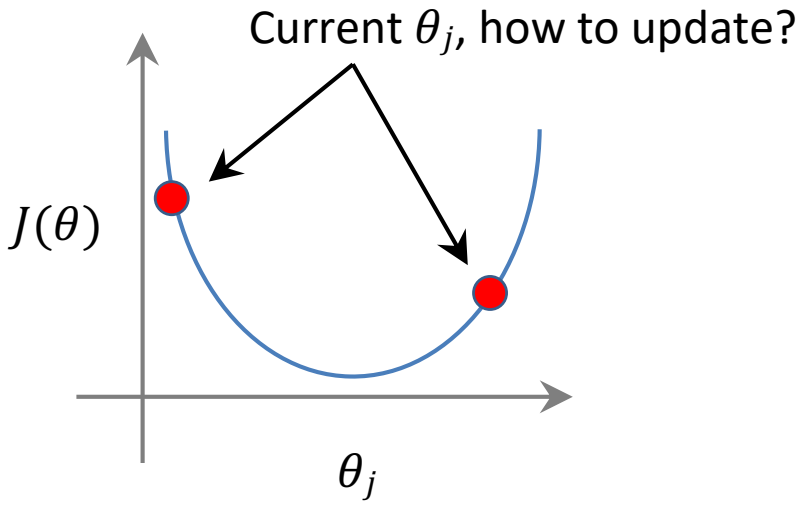
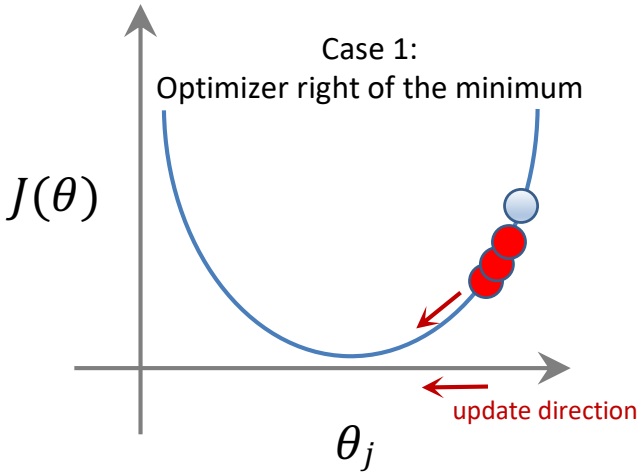


Gradient Descent

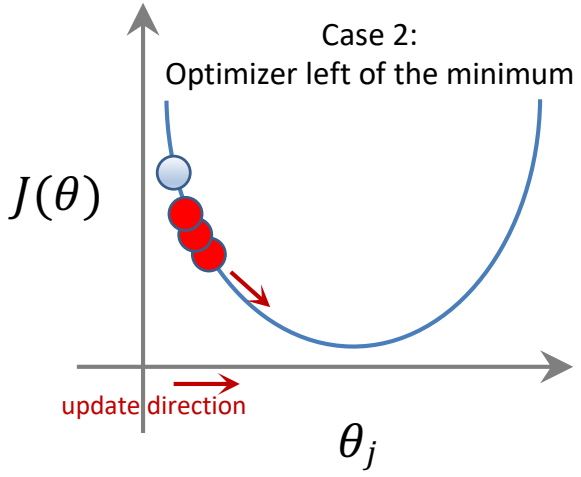


The partial derivative of the loss function is defined as:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \lim_{\varepsilon \to 0} \frac{J(\theta|\theta_j + \varepsilon) - J(\theta|\theta_j)}{\varepsilon} > 0$$

when $\varepsilon \to 0^+$ and $\varepsilon \to 0^-$.

→ To reduce the loss, θ_j must move in the *negative direction* which is opposite to the sign of the partial derivative.



The partial derivative of the loss function is defined as:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \lim_{\varepsilon \to 0} \frac{J(\theta|\theta_j + \varepsilon) - J(\theta|\theta_j)}{\varepsilon} < 0$$

when $\varepsilon \to 0^+$ and $\varepsilon \to 0^-$.

→ To reduce the loss, θ_j must move in the *positive direction* which is opposite to the sign of the partial derivative.

Summary: in both cases, whether $\frac{\partial J(\theta)}{\partial \theta_j} > 0$ or $\frac{\partial J(\theta)}{\partial \theta_j} < 0$, the loss is reduced by moving θ_j in the direction opposite to the sign of its partial derivative, that is, along the negative gradient direction.

Gradient Descent

The Gradient Descent algorithm:

- Choose initial values for θ.
- Until we reach the minimum, update θ to reduce $J(\theta)$:

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^{n} \left(h_\theta\left(x^{(i)}\right) - y^{(i)} \right)^2$$

$$= \frac{\partial}{\partial \theta_j} \frac{1}{2n} \sum_{i=1}^{n} \left(\sum_{k=0}^{d} \theta_k x_k^{(i)} - y^{(i)} \right)^2$$

$$= \frac{1}{n} \sum_{i=1}^{n} \left(\sum_{k=0}^{d} \theta_k x_k^{(i)} - y^{(i)} \right) \times \frac{\partial}{\partial \theta_j} \left(\sum_{k=0}^{d} \theta_k x_k^{(i)} - y^{(i)} \right)$$

$$= \frac{1}{n} \sum_{i=1}^{n} \left(\sum_{k=0}^{d} \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)}$$

Reminder
$(u^2)' = 2uu'$

Gradient Descent

The Gradient Descent algorithm:

- Choose initial values for θ.
- Until we reach the minimum, update θ to reduce $J(\theta)$:

$$\theta_j = \theta_j - \alpha \frac{1}{n} \sum_{i=1}^{n} \left(h_\theta \left(x^{(i)} \right) - y^{(i)} \right) x_j^{(i)}$$

Question: when to stop the algorithm, i.e., when the convergence is considered to happen?

Gradient Descent

The Gradient Descent algorithm:

- Choose initial values for θ.
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$$\theta_j = \theta_j - \alpha \frac{1}{n} \sum_{i=1}^{n} \left(h_\theta \left(x^{(i)} \right) - y^{(i)} \right) x_j^{(i)}$$

Question: when to stop the algorithm, i.e., when the convergence is considered to happen?

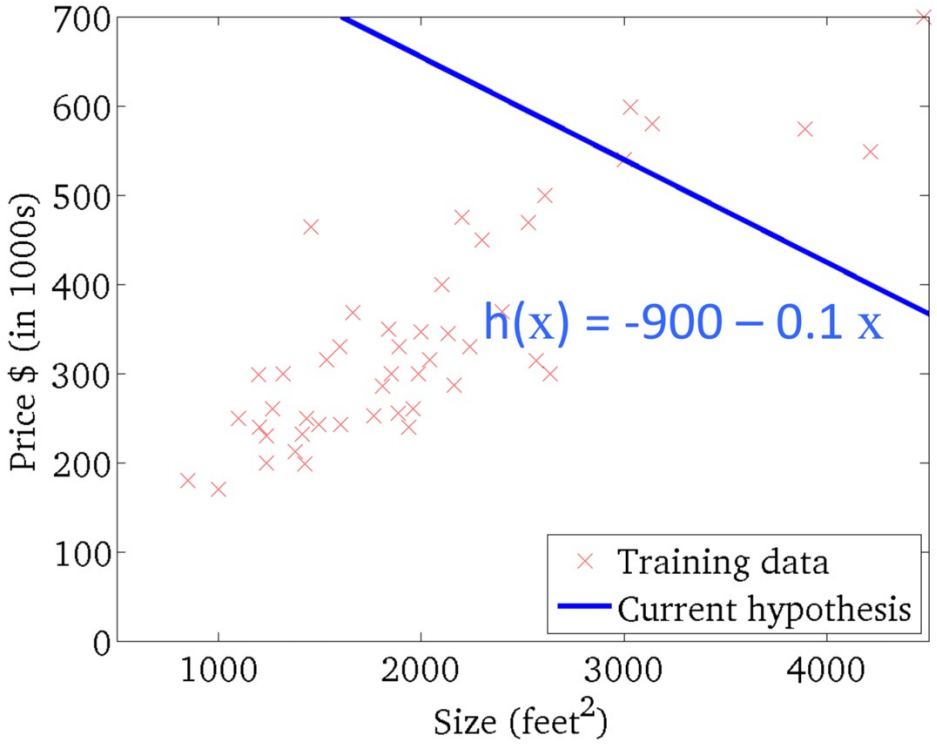
Answer: convergence is reached when either

- $J(\theta) < \varepsilon$
- $\|\theta_{new} - \theta_{old}\|_2 < \varepsilon$ where L2-norm $\|v\|_2 = \sqrt{\sum_i v_i^2} = \sqrt{v_1^2 + v_2^2 + \cdots + v_{|v|}^2}$
- Model accuracy is good enough on an independent test set.

Question: how much is a model accuracy considered to be good enough?

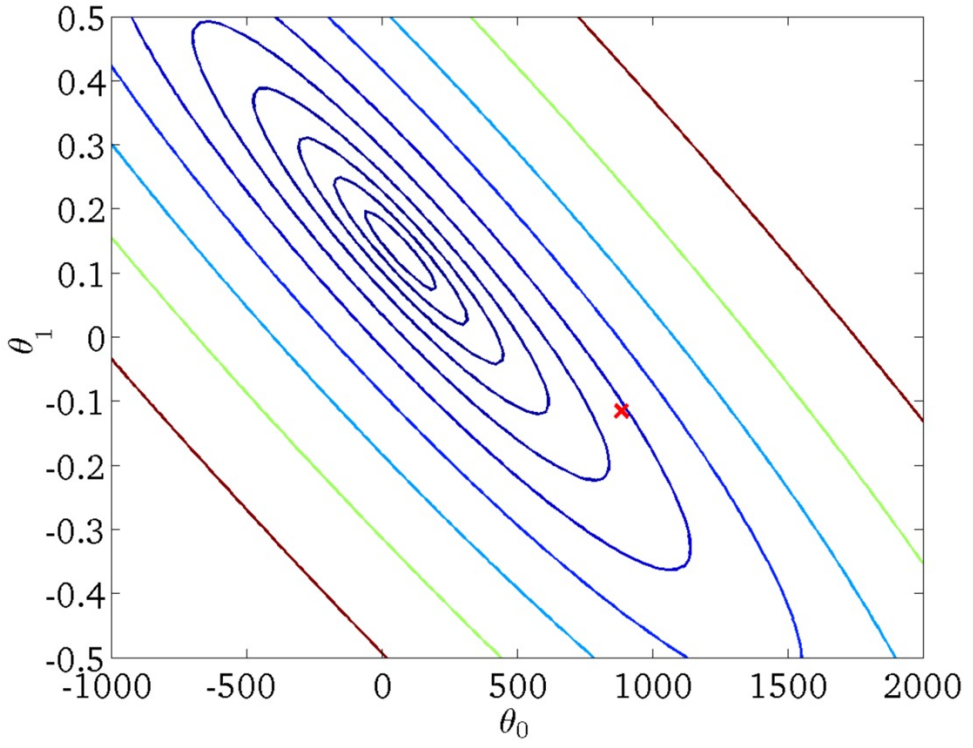
Gradient Descent

$h_\theta(x)$
(for fixed θ_0, θ_1, this is a function of x)



Fitting plot

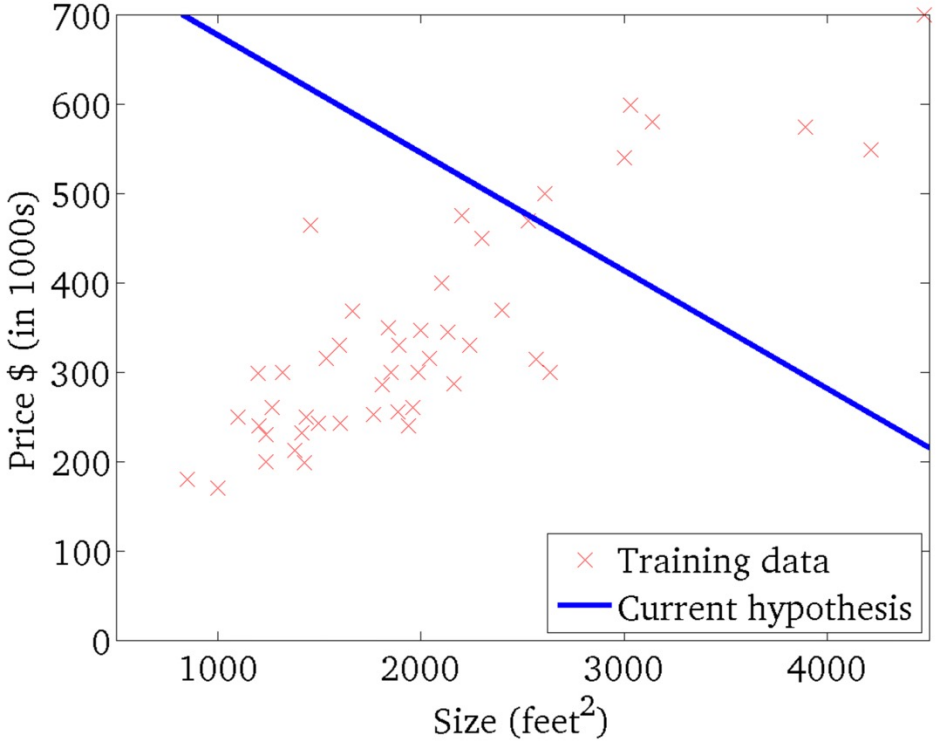
$J(\theta_0, \theta_1)$
(function of the parameters θ_0, θ_1)



Contour plot

Gradient Descent

$h_\theta(x)$
(for fixed θ_0, θ_1, this is a function of x)



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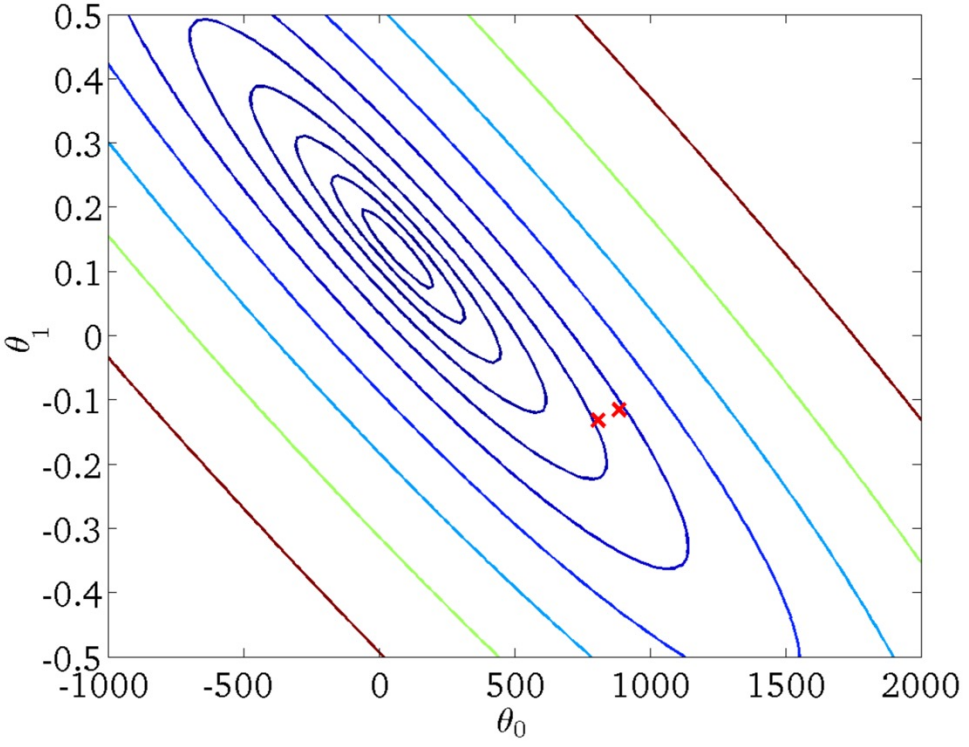

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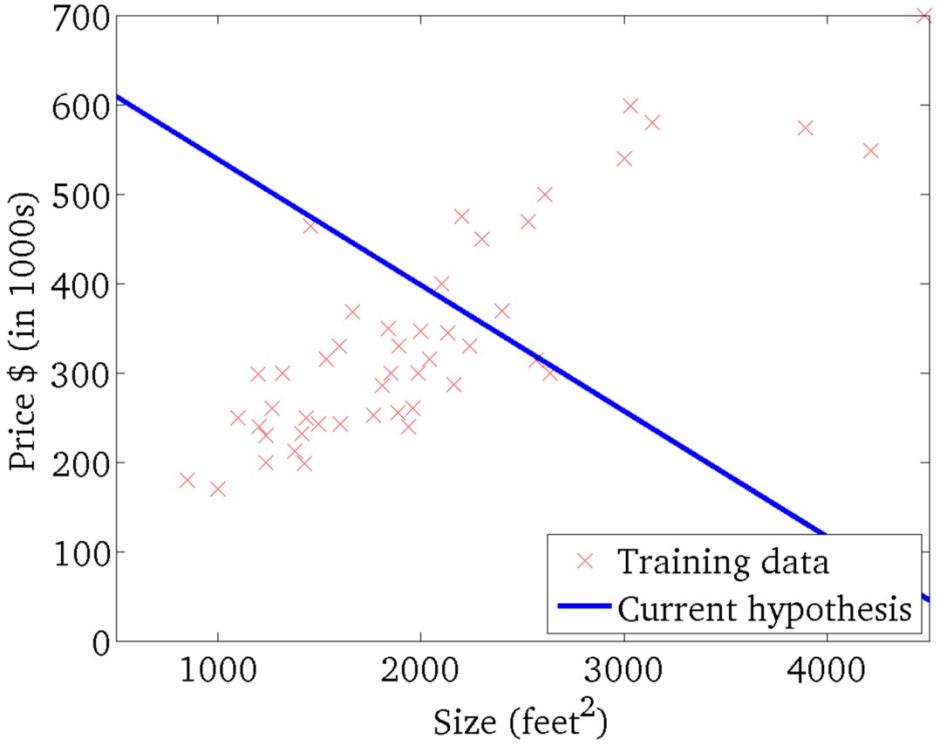

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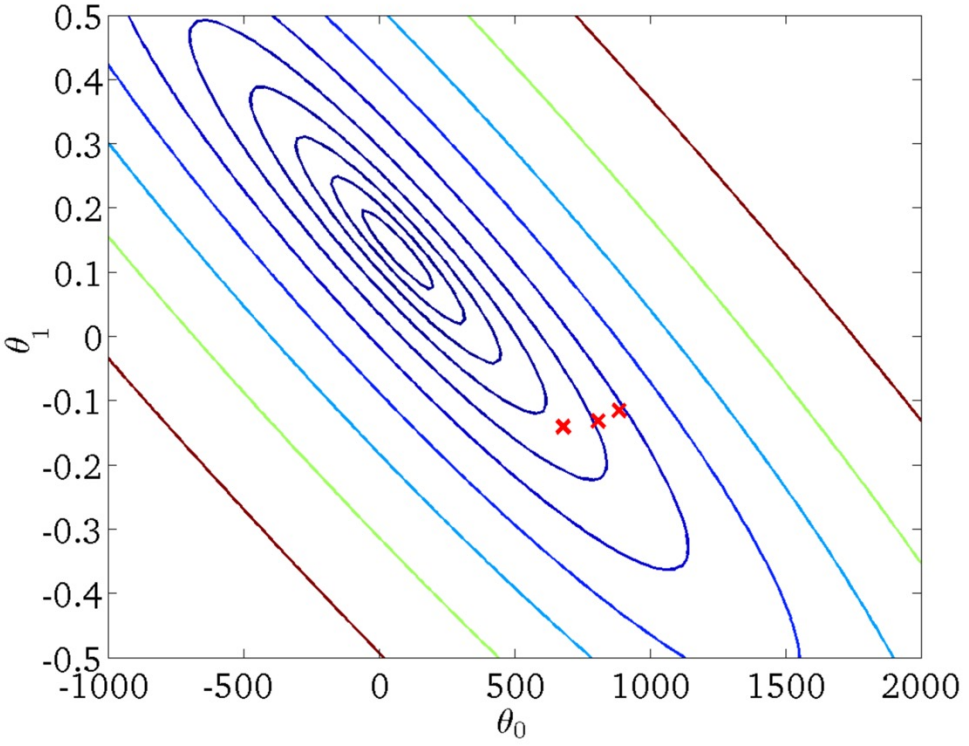

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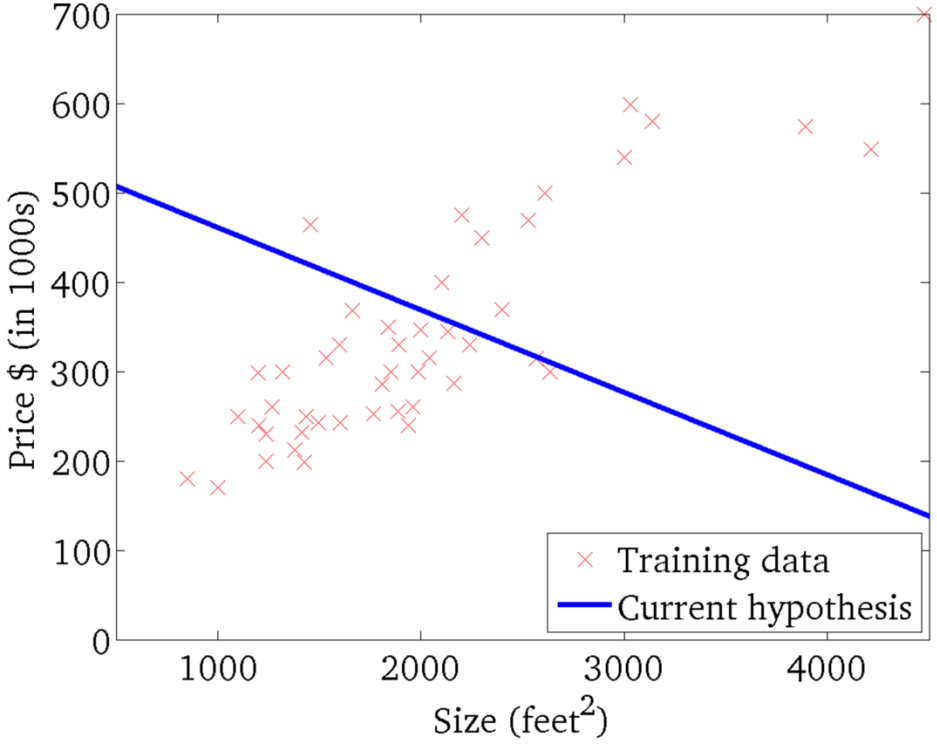
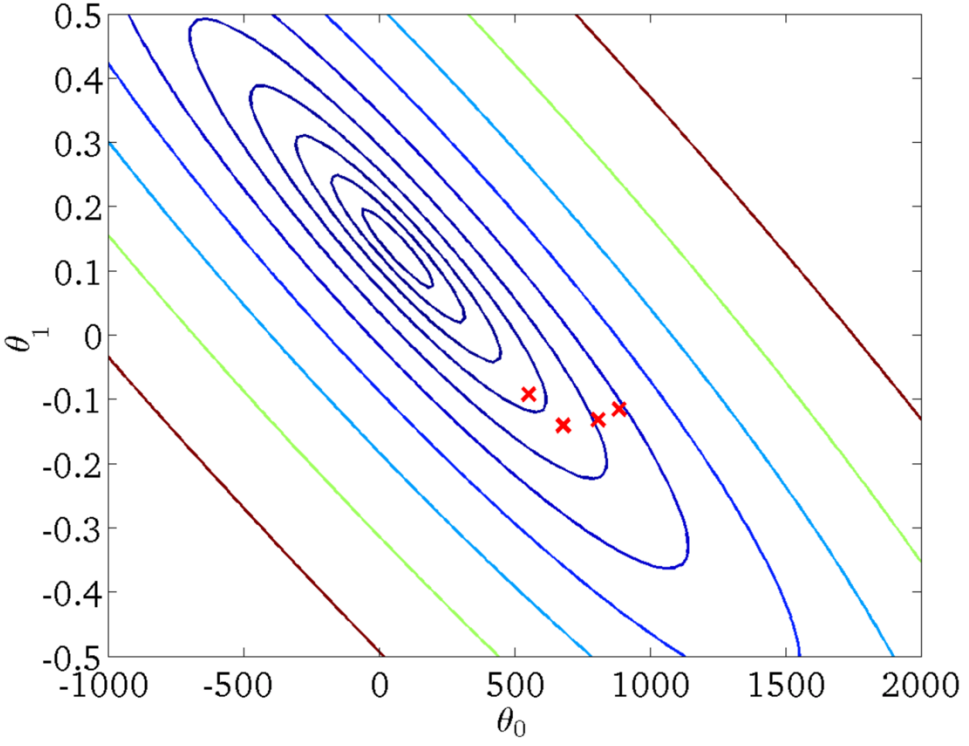


Fitting plot

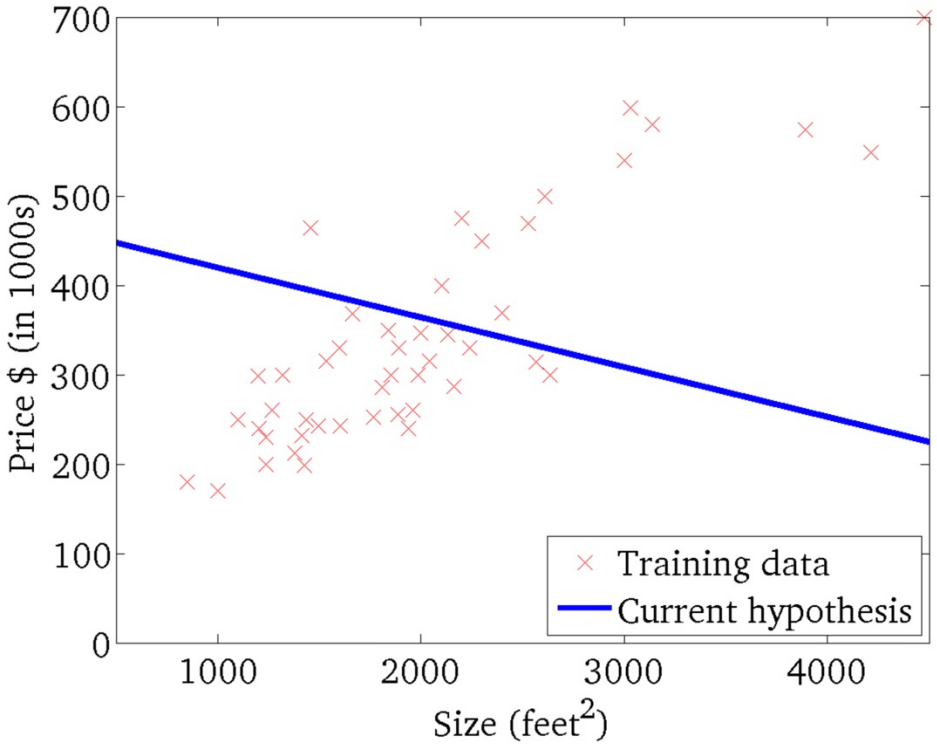
$J(\theta_0, \theta_1)$
(function of the parameters θ_0, θ_1)



Contour plot

Gradient Descent

$h_\theta(x)$
(for fixed θ_0, θ_1, this is a function of x)



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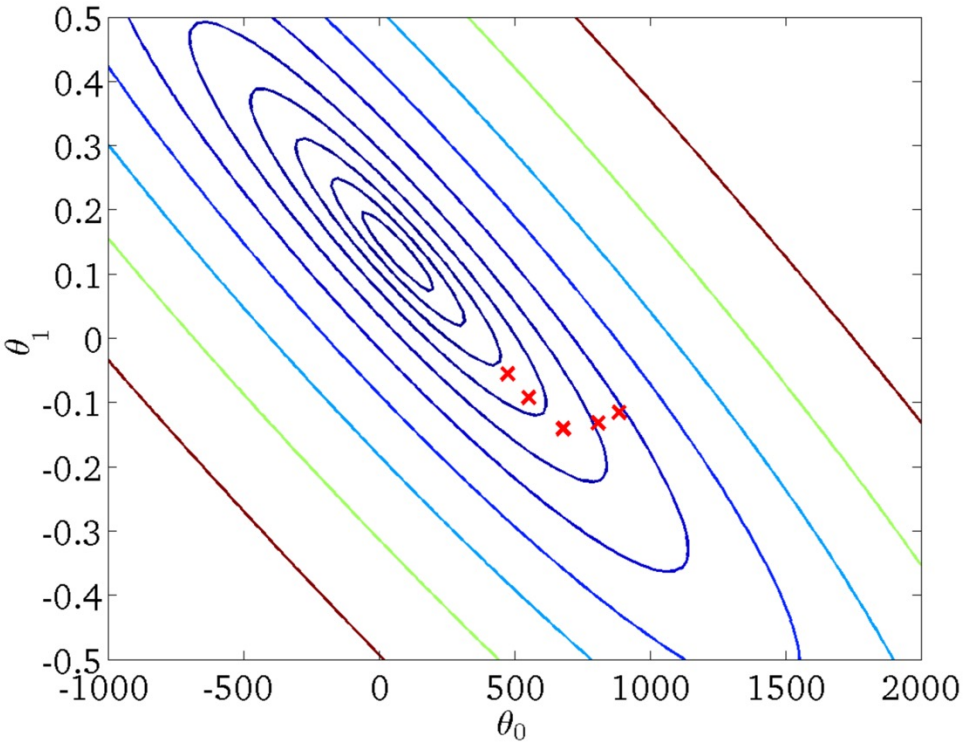

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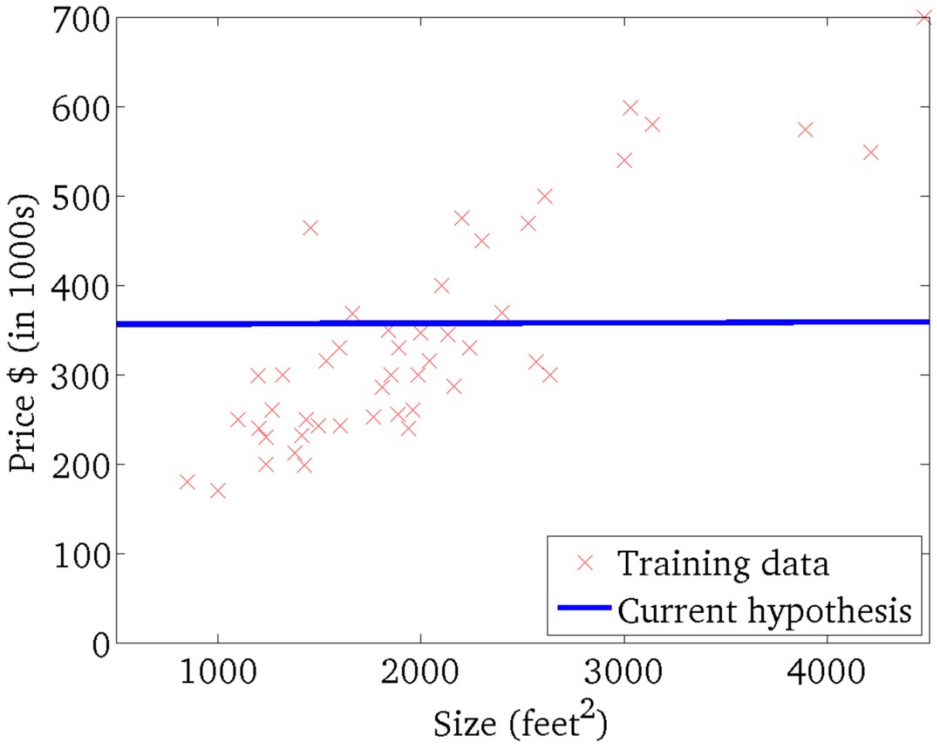

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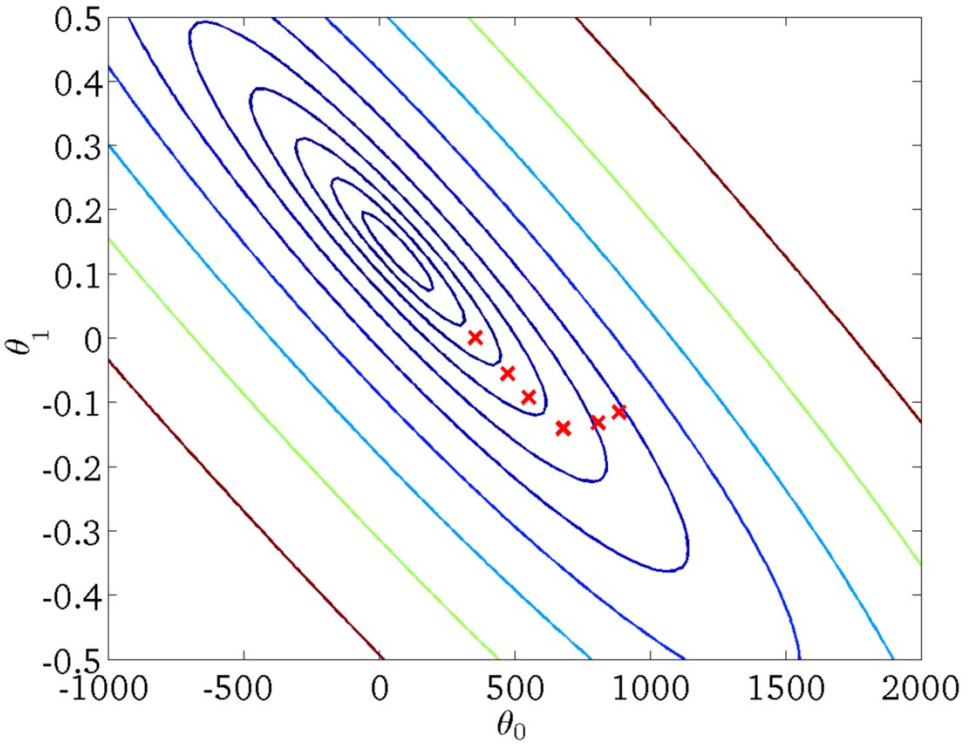

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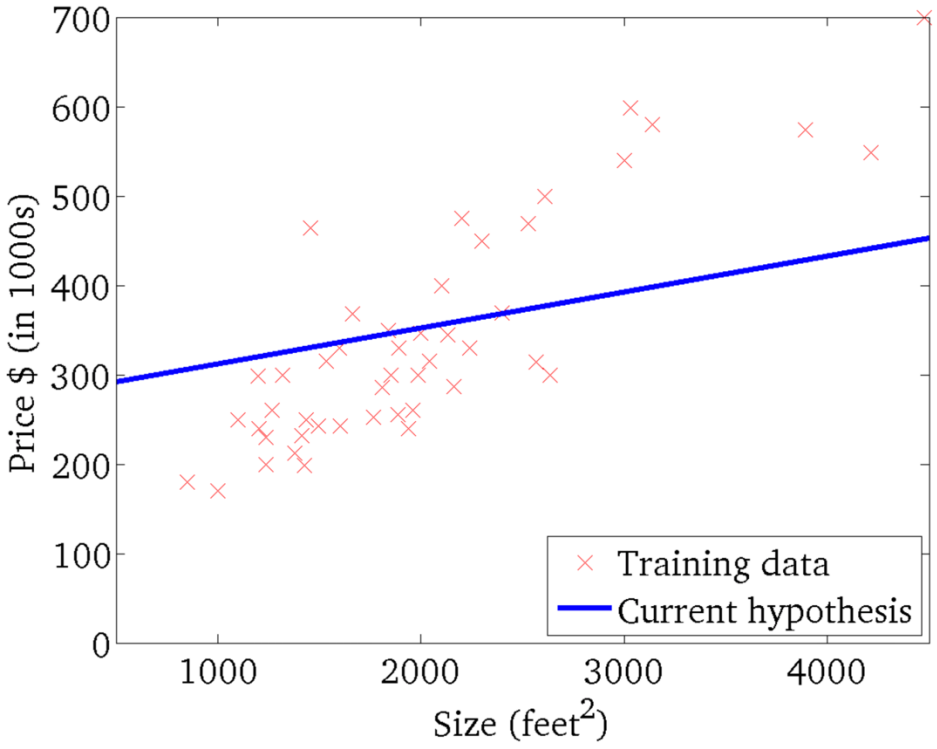

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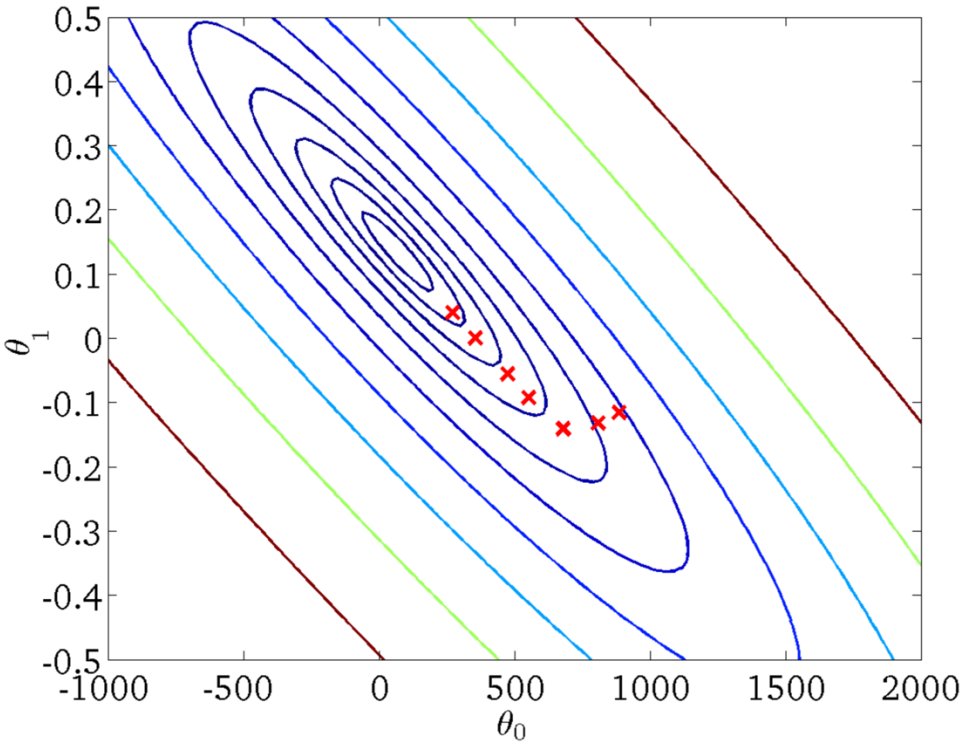

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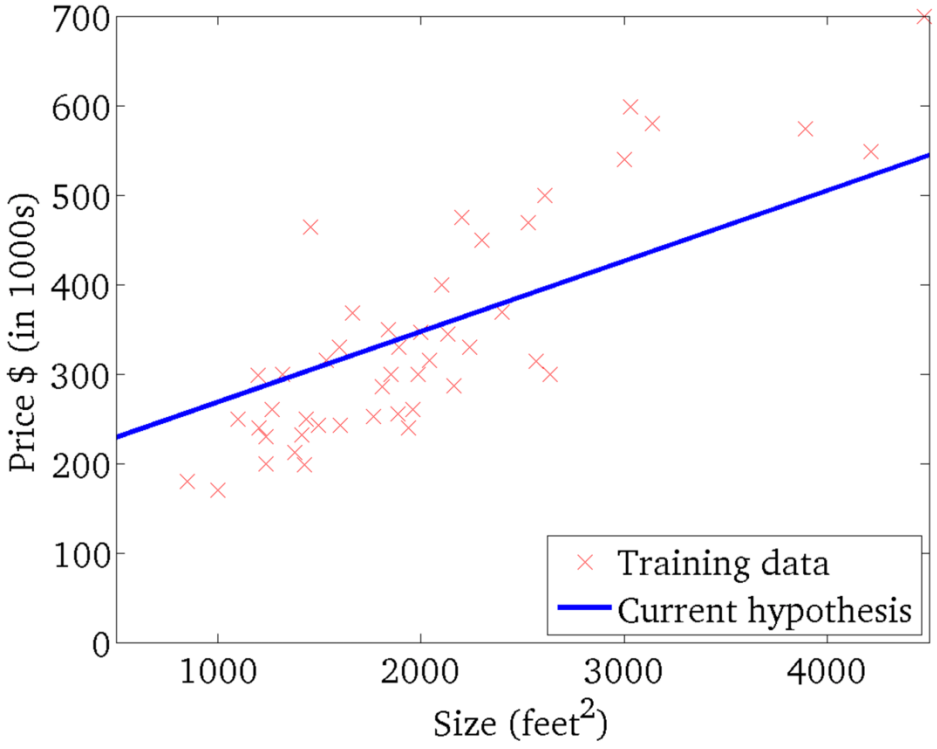

Fitting plot

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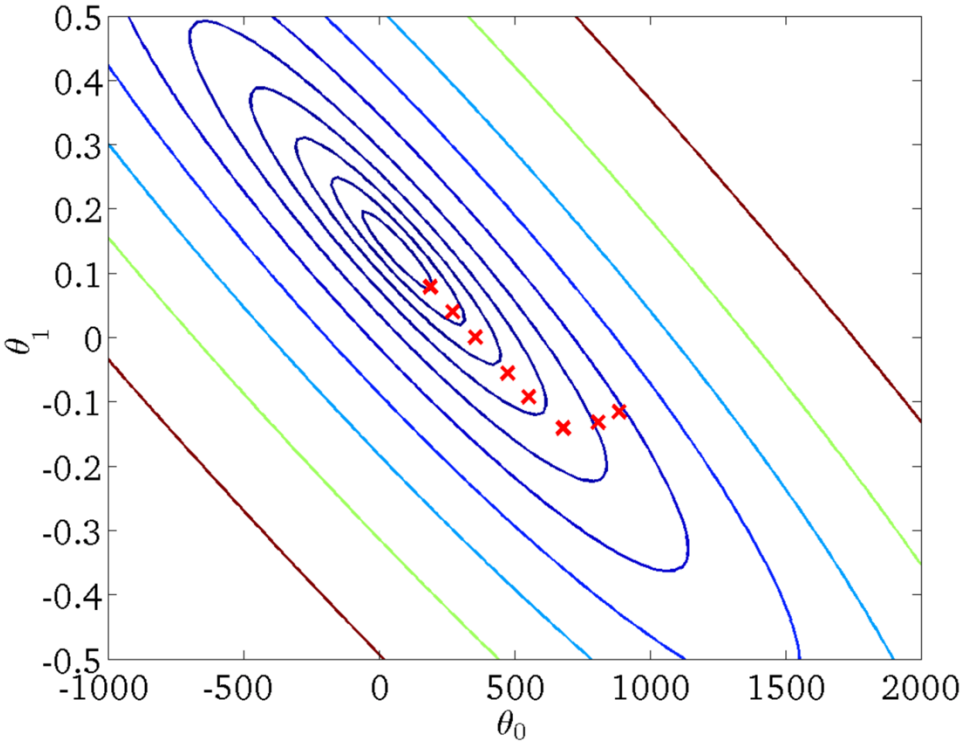

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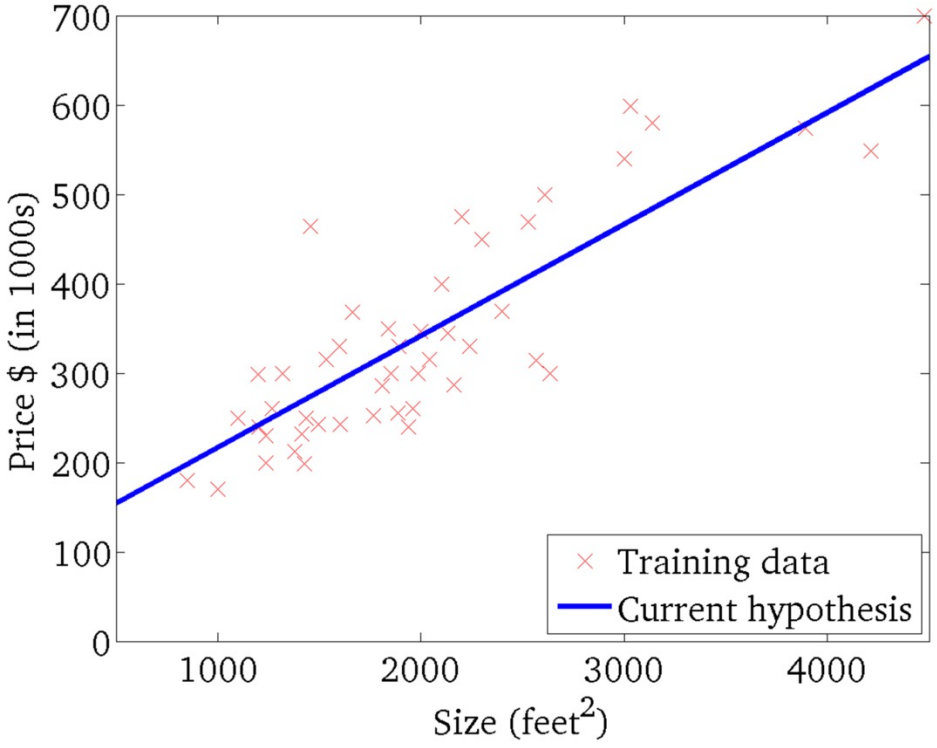

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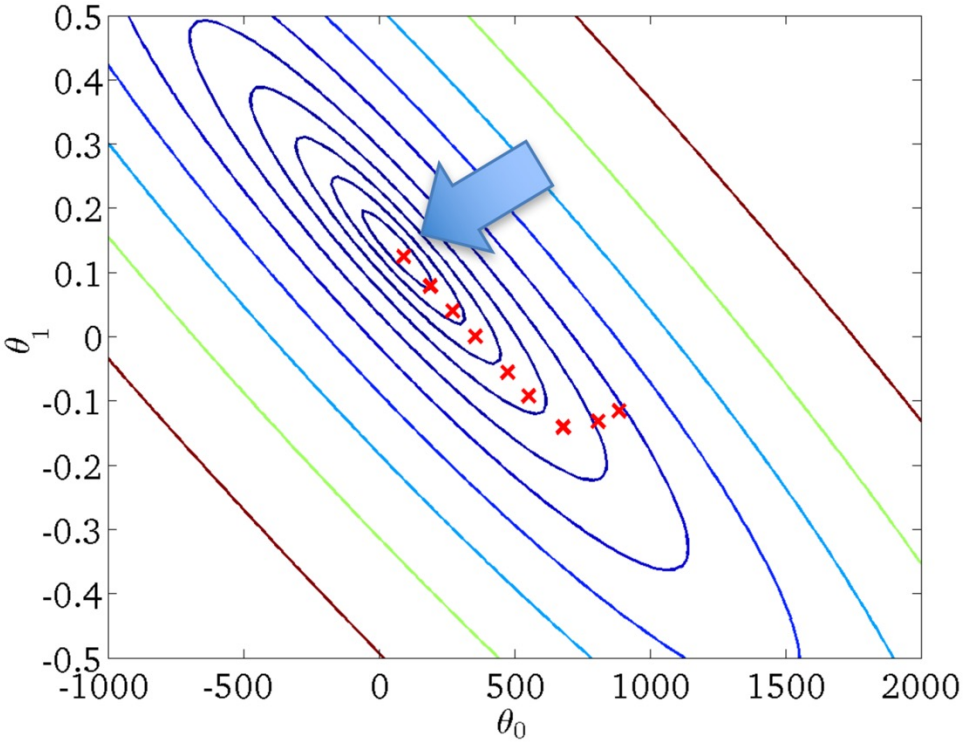


Contour plot

Gradient Descent variants

There are three variants of gradient descent:

- *Batch* gradient descent
- *Stochastic* gradient descent
- *Mini-batch* gradient descent

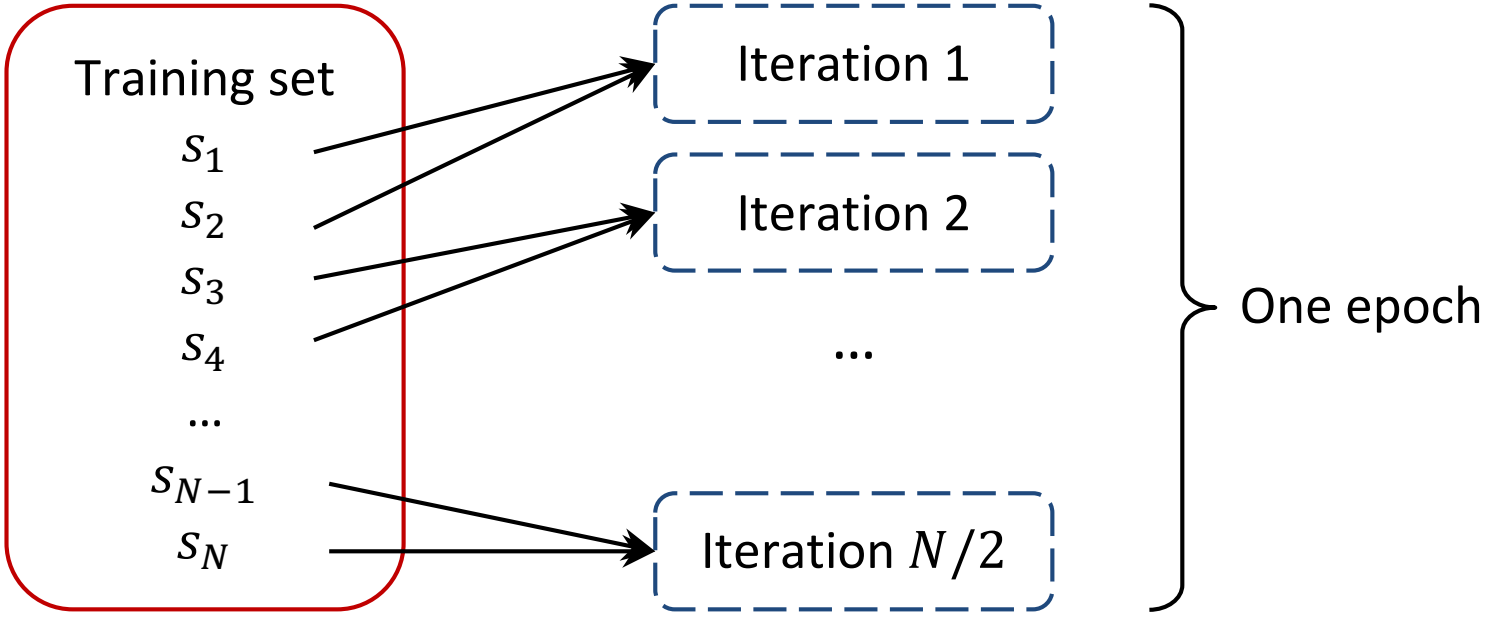
The main difference between these methods is the *amount of data* used to compute the gradient at each update step.

$$\theta_j = \theta_j - \alpha \underbrace{\frac{\partial}{\partial \theta_j} J(\theta)}$$

This amount varies depending on the specific method.

Iteration vs. Epoch

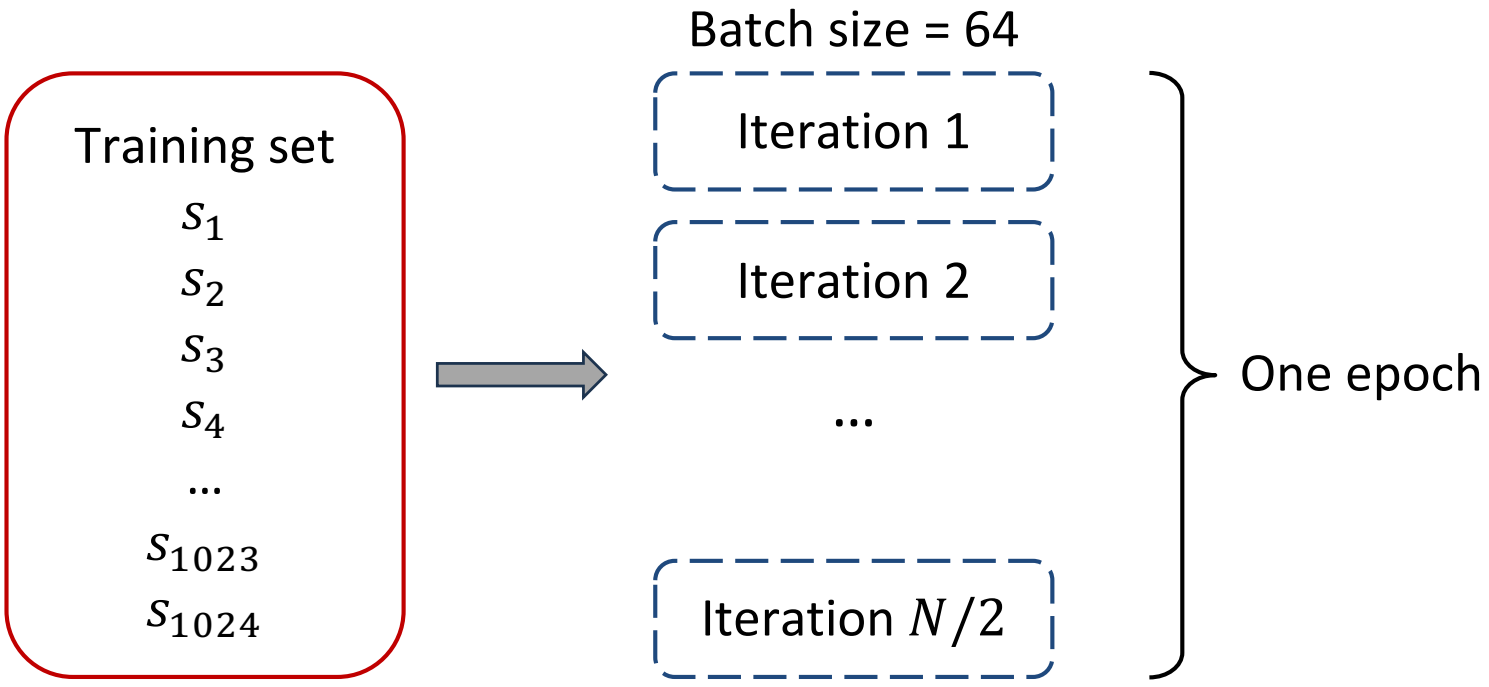
- One iteration corresponds to processing *one batch* of training samples and updating the model once.
- One epoch corresponds to processing the *entire training dataset* once.



An example illustrating the difference between one iteration and one epoch for a dataset with N data samples and batch size of 2.

Iteration vs. Epoch

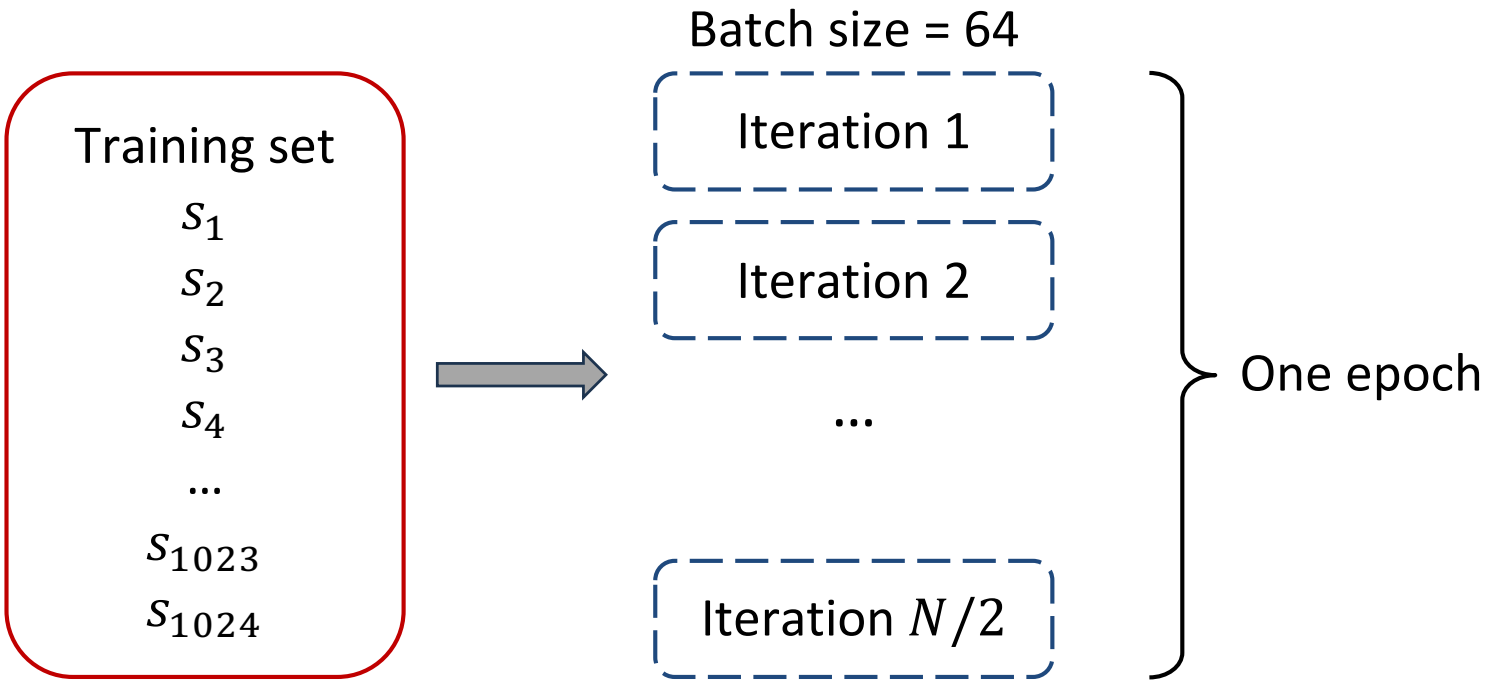
Question: How many iterations are there in one epoch if the dataset has 1024 data samples and the batch size is 64?



Iteration vs. Epoch

Question: how many iterations are there in one epoch if the dataset has 1024 data samples and the batch size is 64?

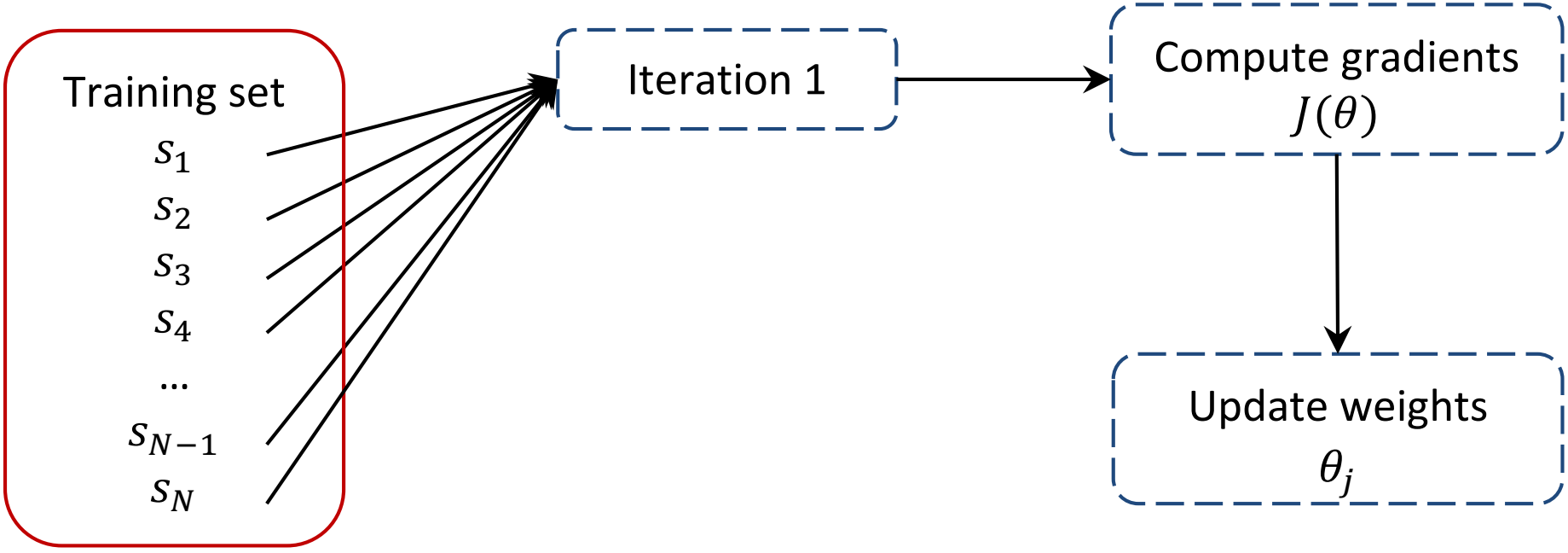
Answer: one epoch is equivalent to 1024/64 = 16 iterations.



Batch gradient descent

Batch gradient descent: Considers the *entire training dataset* at once to calculate the average gradient.

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta, x^{(1:N)}, y^{(1:N)})$$



Batch gradient descent (Batch GD)

Advantages:

- *Accurate updates*: Uses the entire dataset for gradient calculation, leading to a more accurate estimation of the true gradient.
- *Smooth convergence*: Updates tend to be smoother, leading to a more stable descent towards the minimum.

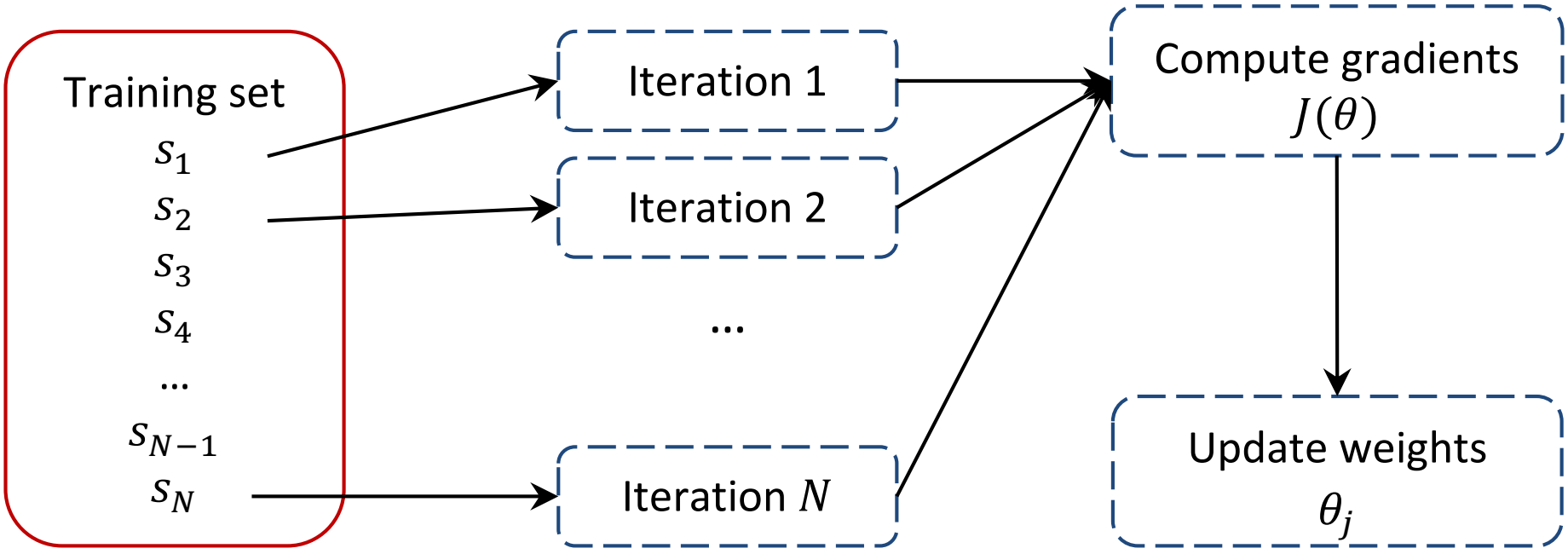
Disadvantages:

- *Slow for large datasets*: This method requires processing the entire dataset during each training iteration, leading to slow training times for big data.
- *Memory Intensive*: Storing the entire dataset in memory is necessary, which can be a challenge for large datasets.

Stochastic gradient descent (SGD)

Stochastic gradient descent: Uses a *single data point* at a time to calculate the gradient.

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta, x^{(i)}, y^{(i)})$$



Stochastic gradient descent

Advantages:

- *Fast:* Processes data one point at a time, making it very fast for large datasets.
- *Less memory intensive:* Only needs to store a single data point in memory at a time.
- *Can escape local minima:* Noise from individual examples can sometimes help escape local minima.

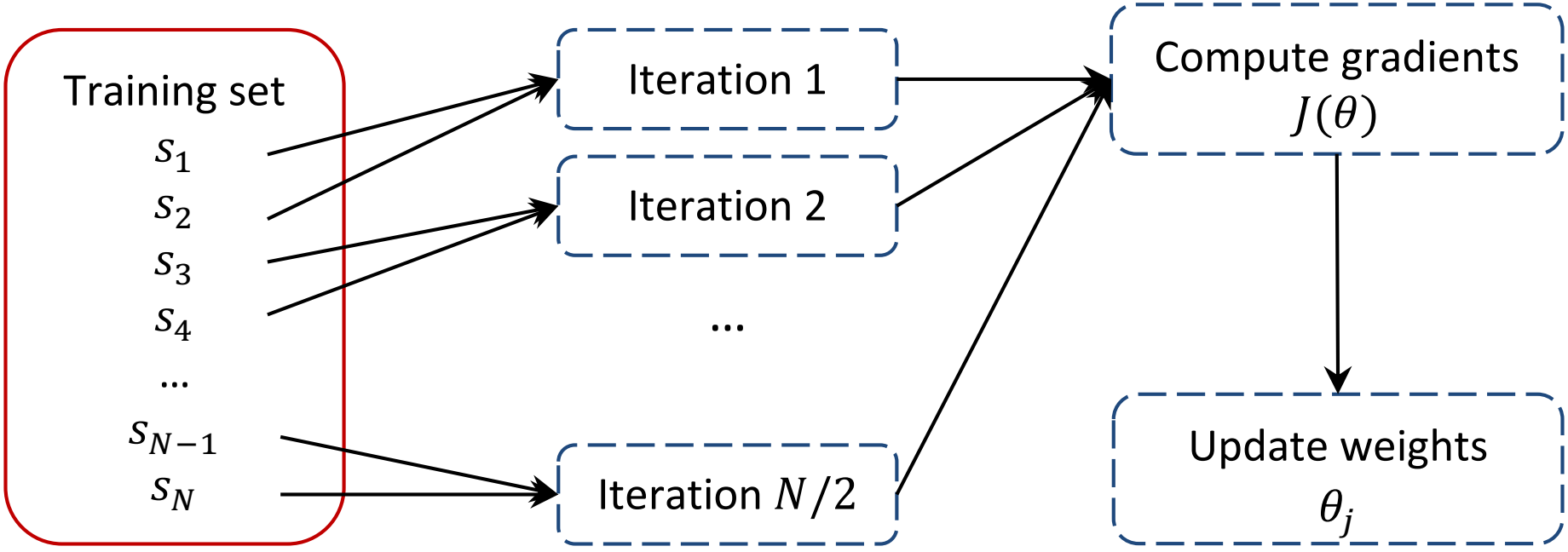
Disadvantages:

- *Noisy updates:* Updates based on single data points are noisy, leading to erratic convergence and oscillations.
- *Less stable convergence:* Can be less stable than Batch GD, potentially getting stuck in bad local minima.

Mini-batch gradient descent

Mini-batch gradient descent: Computes the gradient using a small subset of the training data, called a mini batch. The mini batch size k is a hyperparameter that can be adjusted.

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta, x^{(i:i+k)}, y^{(i:i+k)})$$



Mini-batch gradient descent

Advantages:

- *Faster than Batch GD:* Processes data in batches, making it faster for large datasets.
- *More Stable than SGD:* Uses more data than SGD, leading to smoother updates and less noisy convergence.
- *Less Memory Intensive:* Requires storing only a small batch of data in memory.

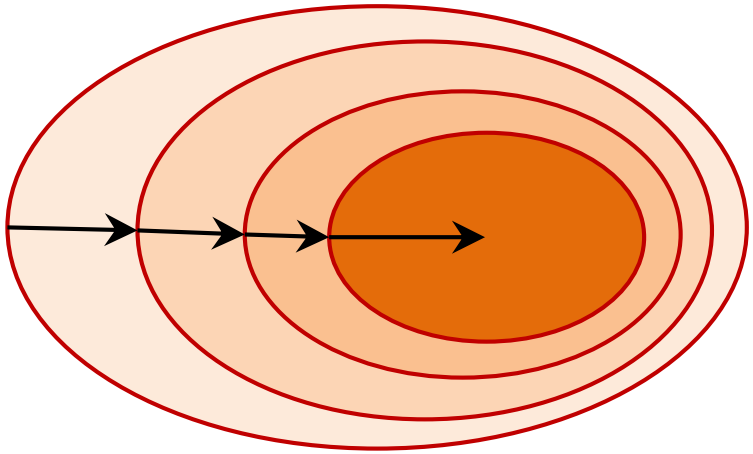
Disadvantages:

- *Not as Accurate as Batch GD:* Updates are less accurate than Batch GD because they only use a subset of the data.
- *Tuning Batch Size:* Requires choosing the right batch size for optimal performance.

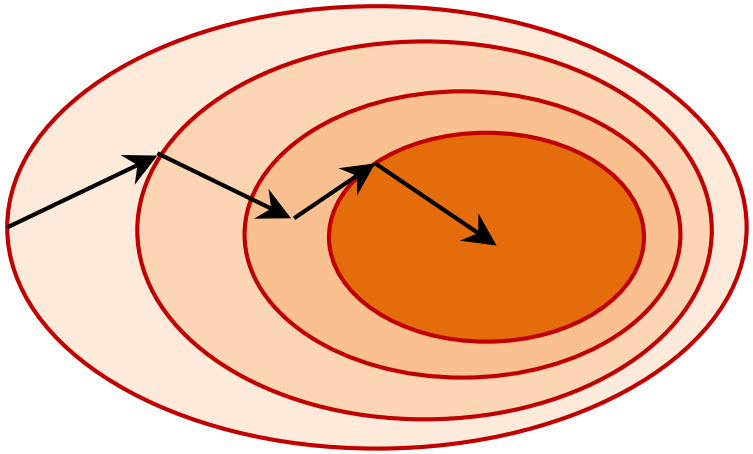
Question: How to choose the optimal batch size?

Batch vs. Mini-batch vs. Stochastic gradient descent

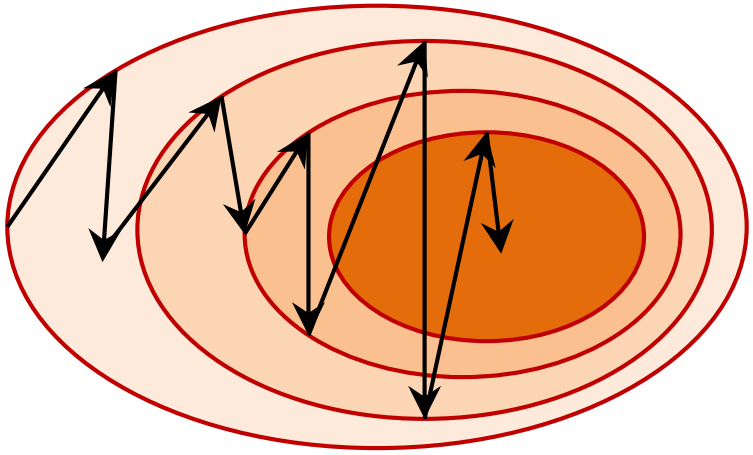
Batch gradient descent



Mini-batch gradient descent



Stochastic gradient descent



Comparison of convergence behavior in Batch, Mini-batch, and Stochastic Gradient Descent

Summary of gradient descent variants

Comparison of Batch, Mini Batch, and Stochastic gradient descent in terms of convergence quality, computational time, and memory consumption.

Method	Convergence	Running time	Memory Usage
Batch gradient descent	Good	Slow	High
Stochastic gradient descent	Not very good	Fast	Low
Mini-batch gradient descent	Medium	Medium	Medium

Note: With the use of GPUs, Batch and Mini-batch gradient descent can handle multiple data samples simultaneously, leading to faster running time.

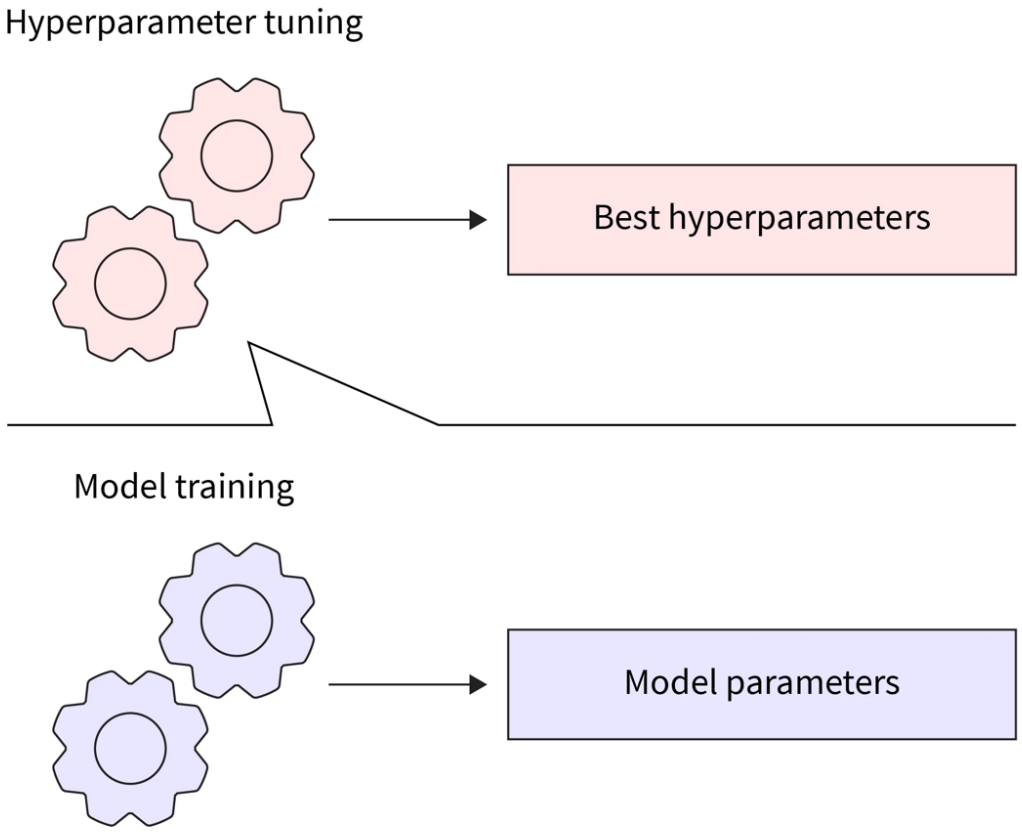
Question: Which gradient descent variant is most widely used in real world applications?

Model hyper-parameters

In machine learning, a hyperparameter is a setting that you choose before training starts. It controls how the model learns, instead of being estimated from the data.

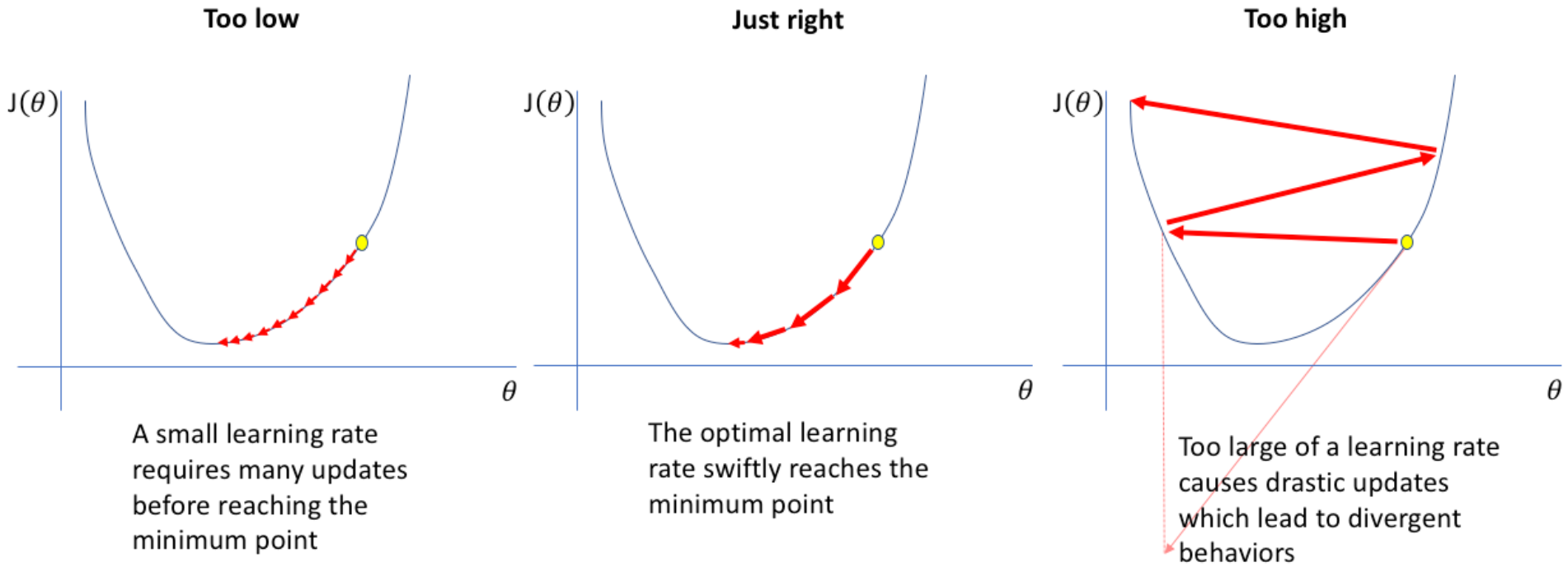
Examples of hyperparameters:

- Batch size
- Learning rate
- Model architecture or function
- Regularization weight
- And others



Model hyperparameters - Learning Rate

Learning rate is a hyperparameter. It determines the step size used to update the model's learnable parameters during training.



Impact of learning rate on model convergence in three scenarios: too low, just right and too high learning rate.

Question: How can we know when to increase / decrease / keep the same the learning rate when training a model?

Linear Regression for more complex models

Linear regression can have complex forms:

- Transformation of quantitative inputs:

  Example: $h_\theta(x) = \theta_0 + \theta_1 \log(x_1) + \theta_2 x_2^2 + \theta_3 \sqrt{x_3}$

- Polynomial transformation

  Example: $h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2^2 + \theta_3 x_3^3$

- Interaction between variables:

  Example: $h_\theta(x) = \theta_0 + \theta_1 x_1 x_2 + \theta_2 x_1 x_3 + \theta_d x_2 x_3$

The general form of the transformations is:

$$h_\theta(x) = \sum_{j=0}^{d} \theta_j \underbrace{\emptyset_j(x)}_{\text{Basis function}}$$

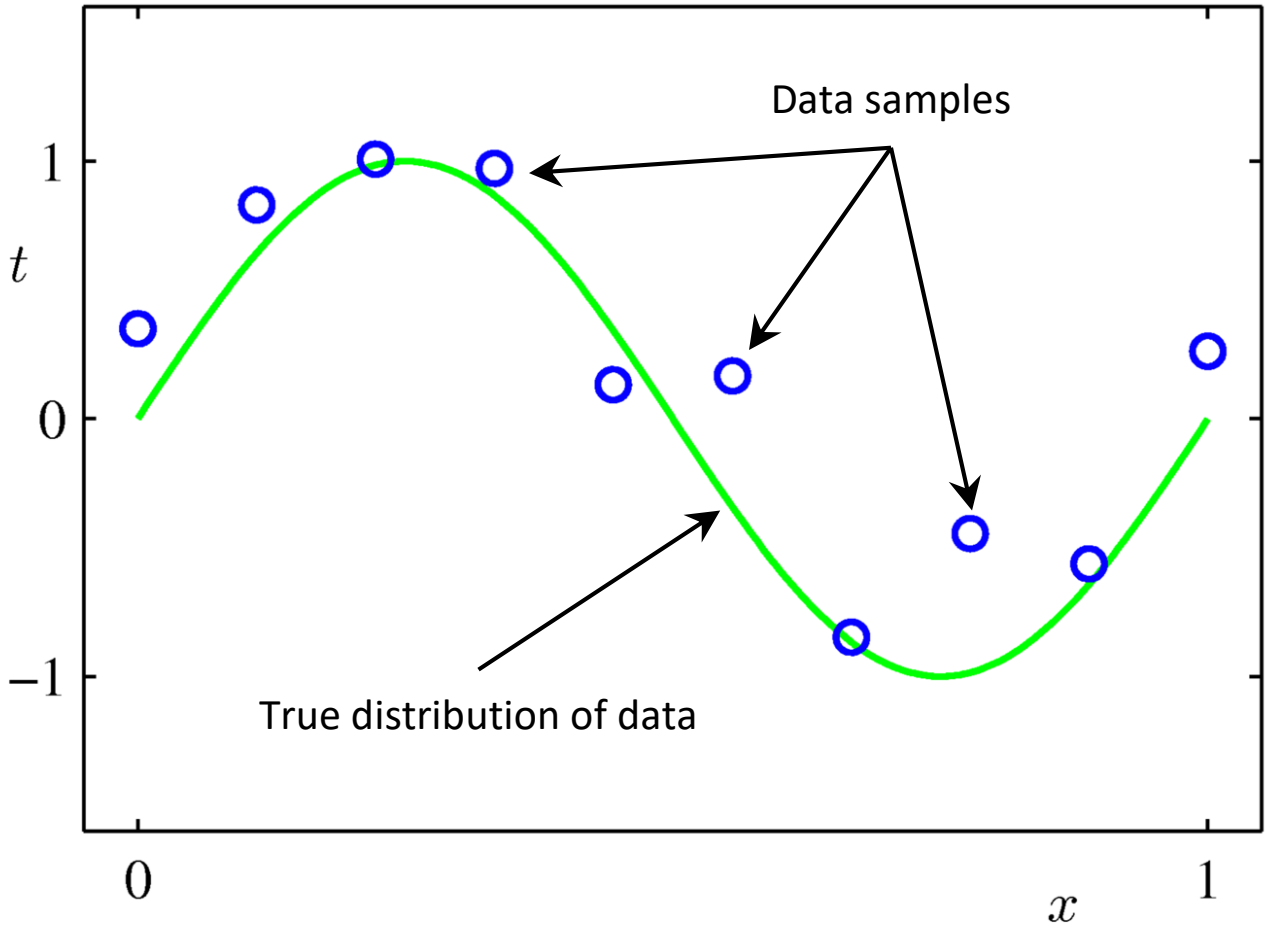
where typically $\emptyset_0(x) = 1$ so that θ_0 acts as a bias.
In the simplest case, we use linear basis function:

$$\emptyset_j(x) = x_j$$

These transformation allows the use of linear regression to fit non-linear dataset.

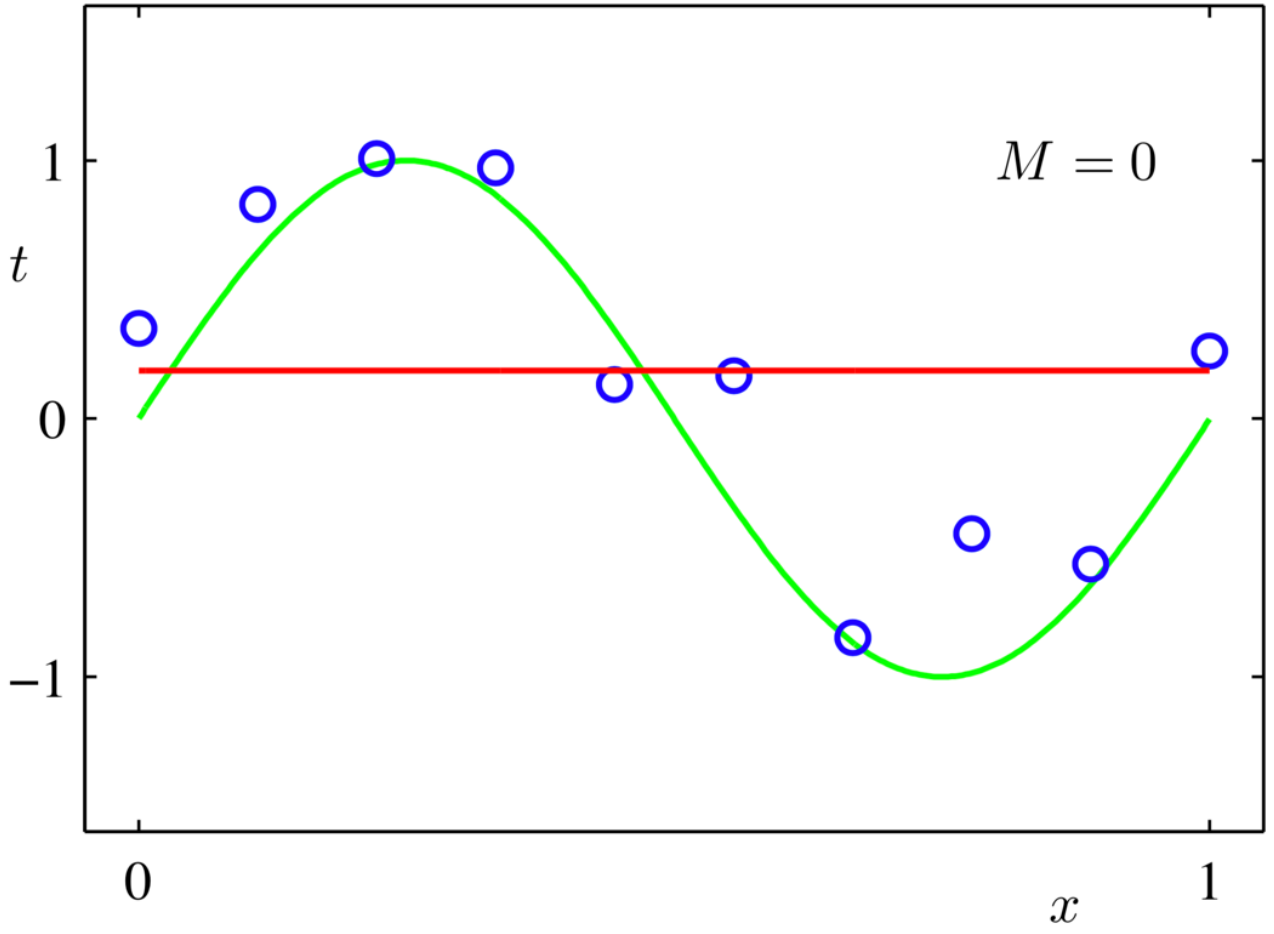
Question: The transformations are not linear. Does linear regression still work/apply?

Example of fitting a Polynomial Curve with Linear Regression



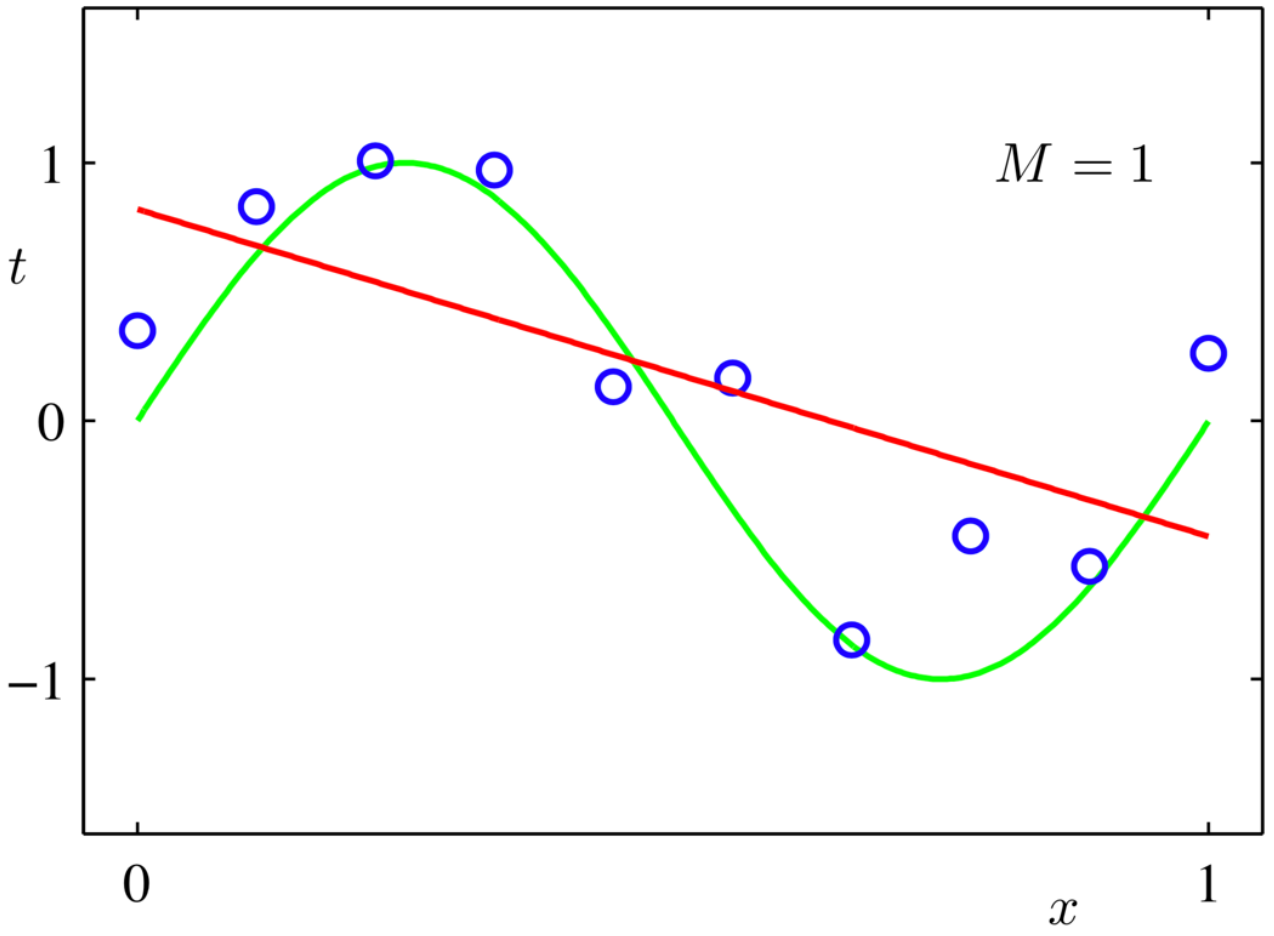
$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_d x^d = \sum_{j=0}^{d} \theta_j \, x^j$$

Example of fitting a Polynomial Curve with Linear Regression



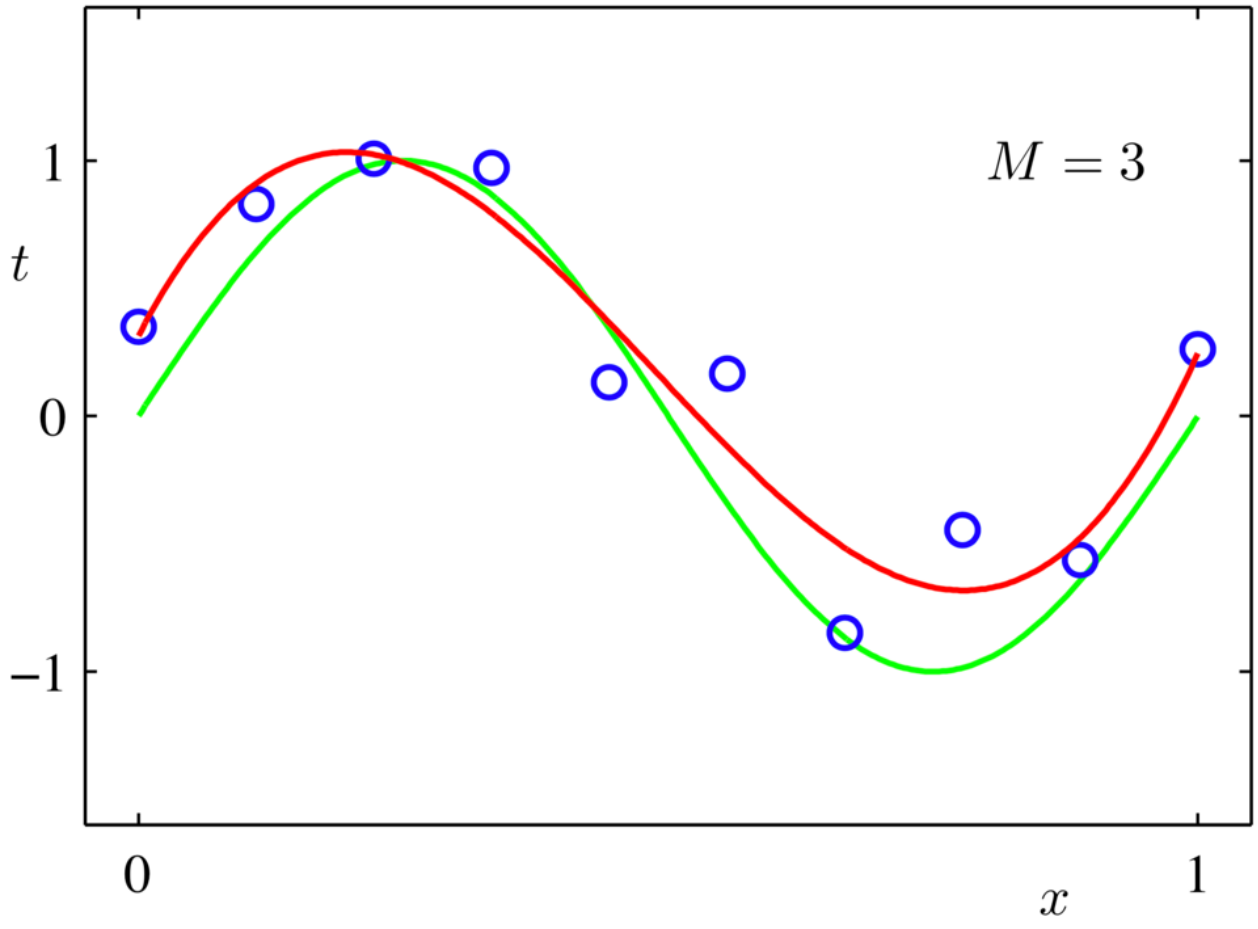
$$h_\theta(x) = \theta_0$$

Example of fitting a Polynomial Curve with Linear Regression



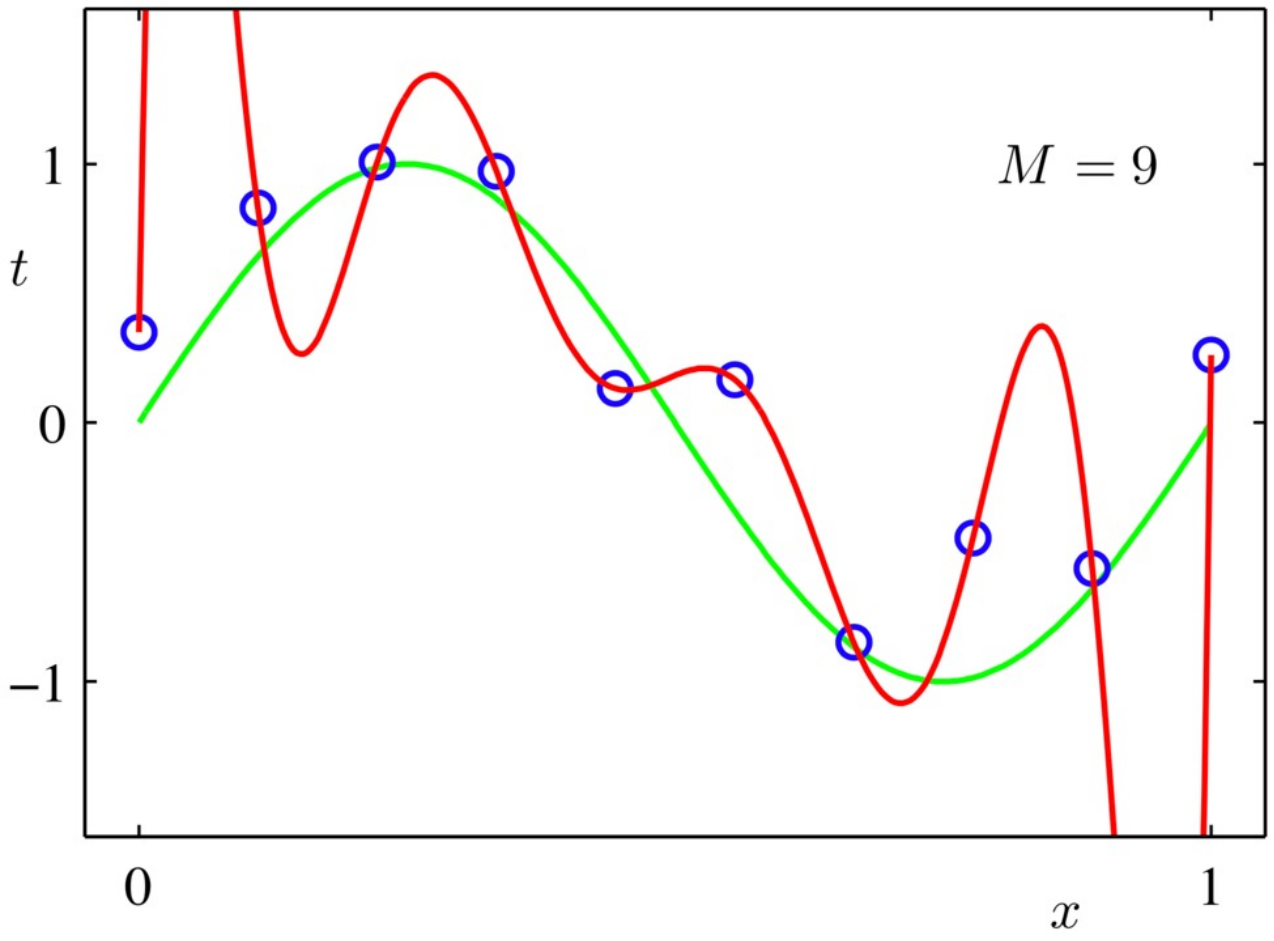
$$h_\theta(x) = \theta_0 + \theta_1 x$$

Example of fitting a Polynomial Curve with Linear Regression



$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

Example of fitting a Polynomial Curve with Linear Regression

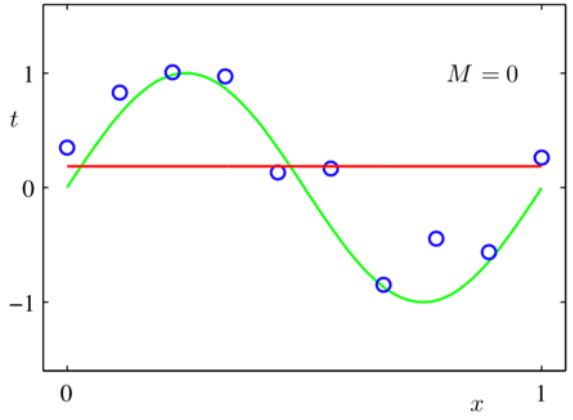


$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \cdots + \theta_9 x^9$$

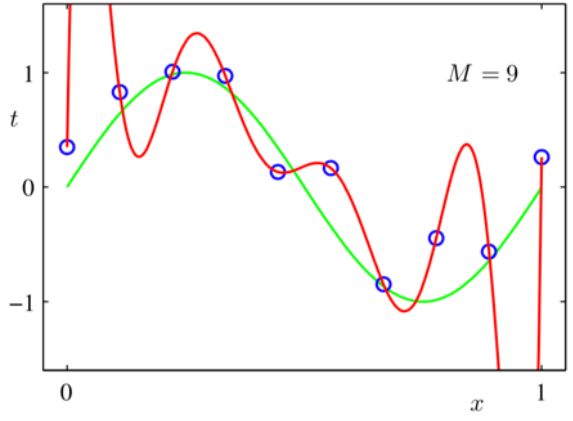
Question: Is the model (the red line) a good fit? Why?

Underfitting and Overfitting

Underfitting: when the model is too simple

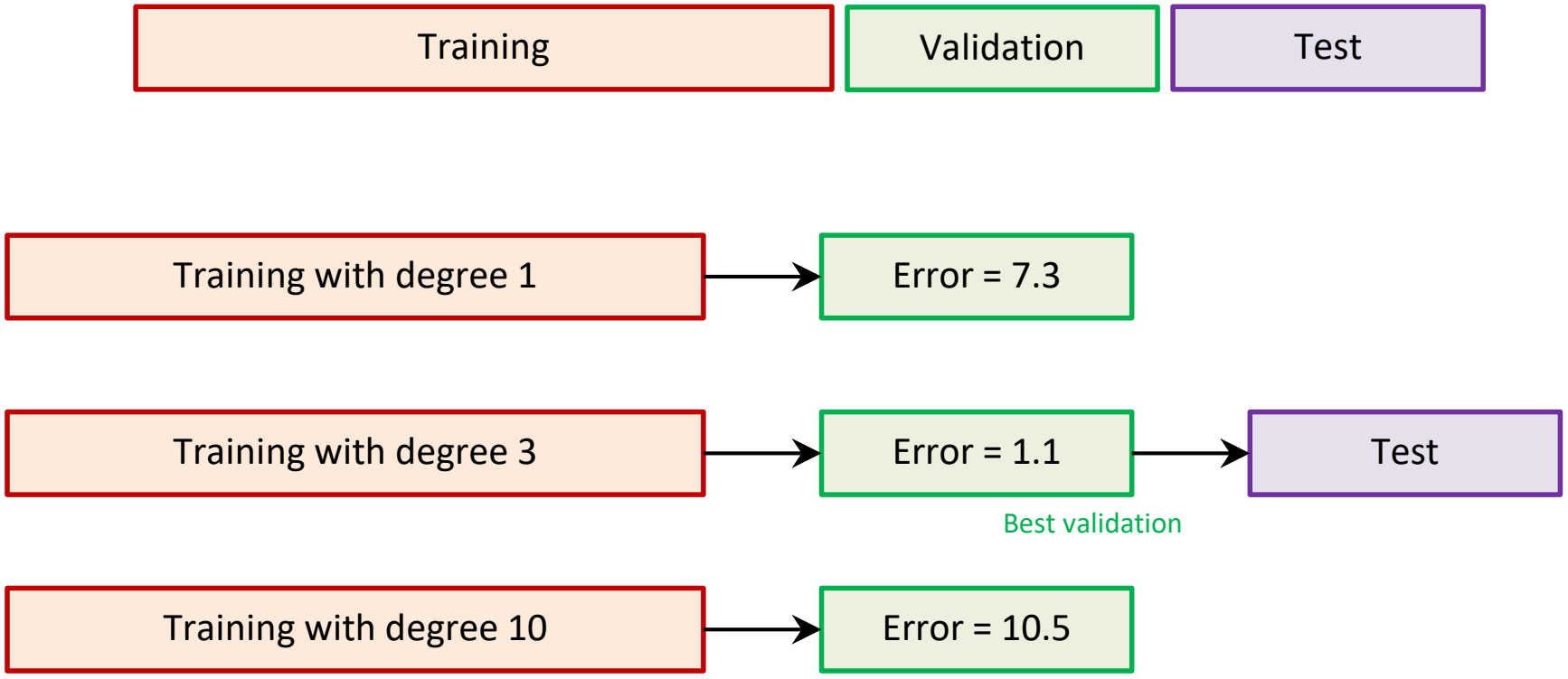


Overfitting: when the model is too complex



Model generalization – Training, validation and test split

Models should be generalized to data they have not seen before.

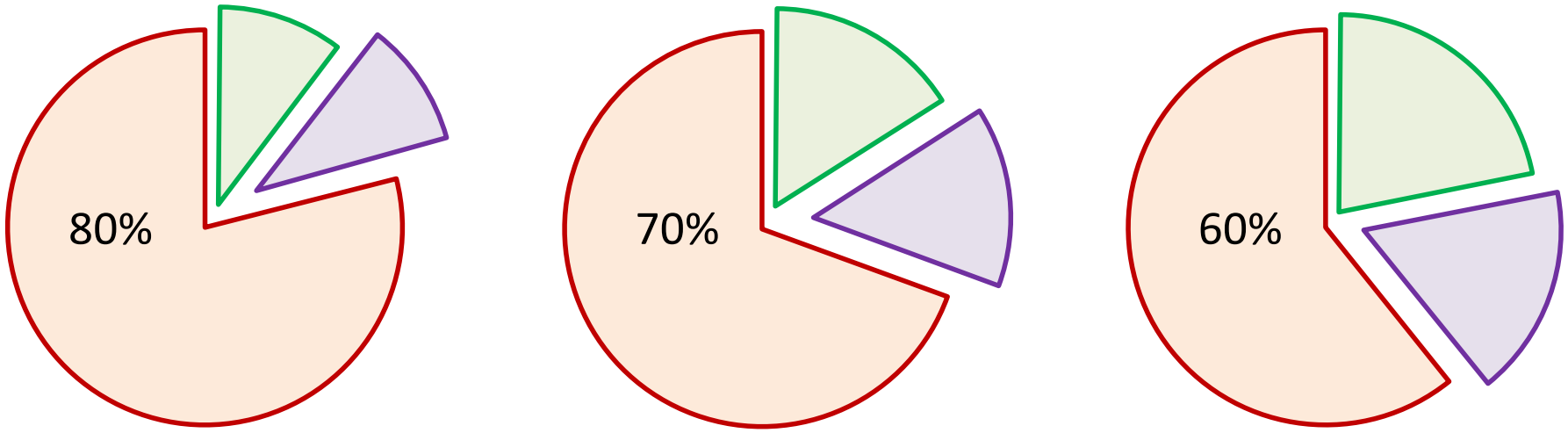


Example of how to train, validate and test the models to ensure generalization.

Question: Question: What is a good train, validation, test split ratio?

Model generalization – Training, validation and test split

Question: Question: What is a good train, validation, test split ratio?



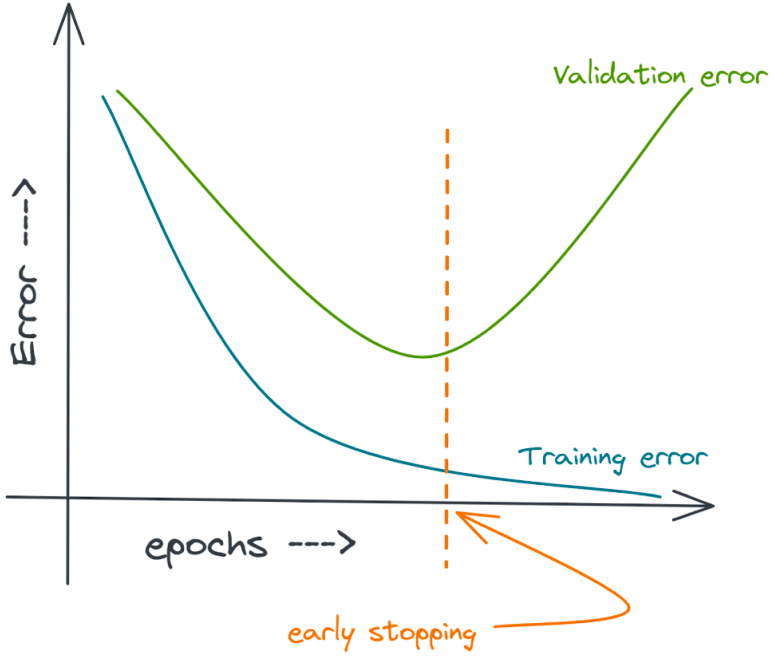
Training set – Validation set – Test set

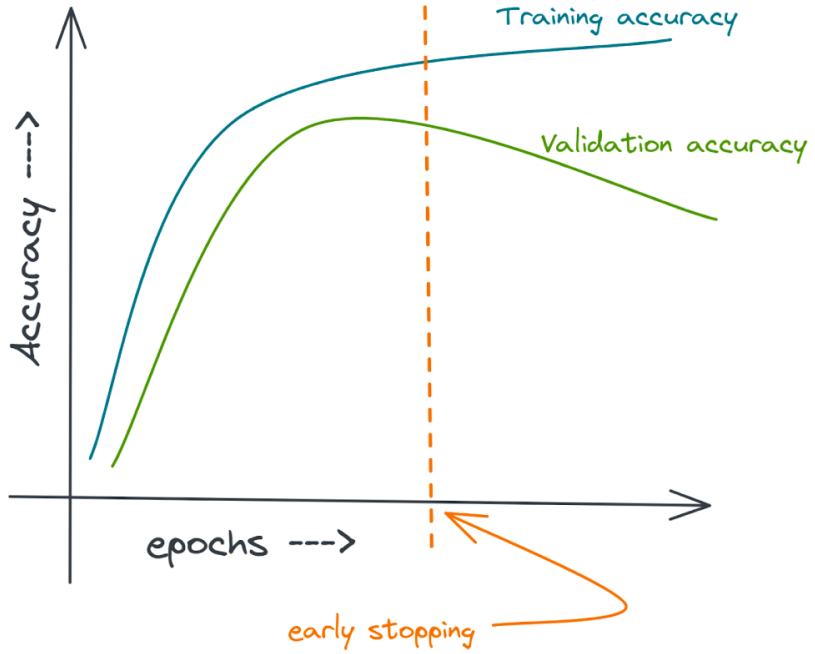
Question: Observing the model's accuracy solely on the training set might result in overfitting. With the use of validation set, how can we determine the right moment to halt the training process?

Model generalization – Training, validation and test split

Question: Observing the model's accuracy solely on the training set might result in overfitting. With the use of validation set, how can we determine the right moment to halt the training process?

Answer: By looking at the model performance on validation set.

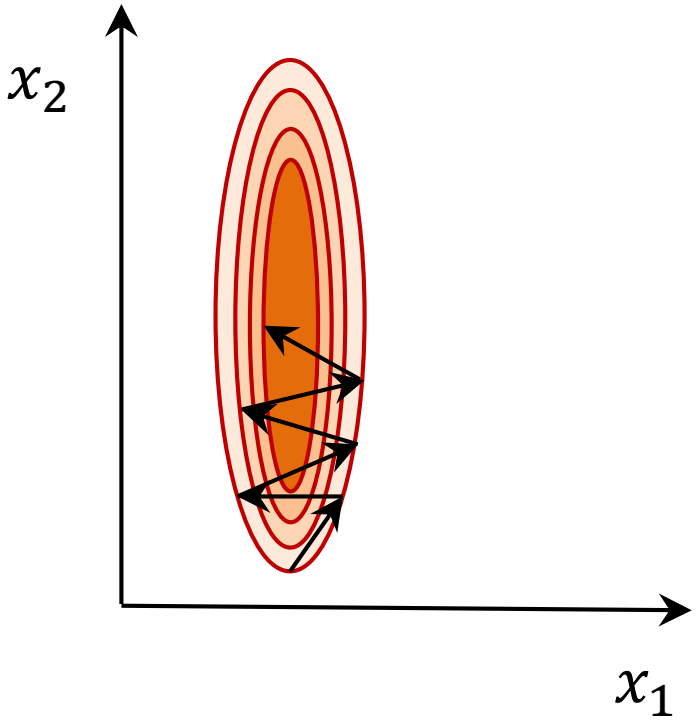




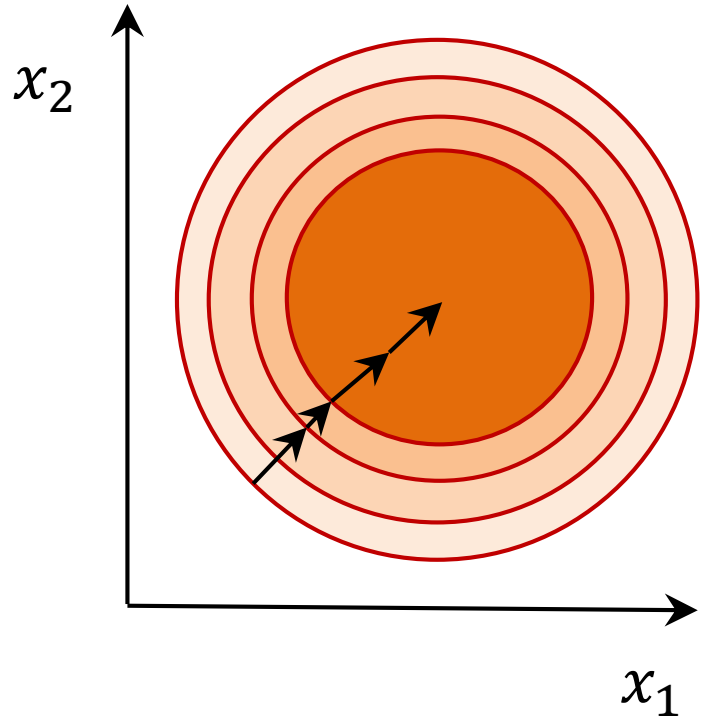
Feature Scaling

Feature scaling is a preprocessing technique used in machine learning to standardize the range of independent variables or features in the dataset.

Before feature scaling



After feature scaling



Feature Scaling – Standardization

Standardization rescales features to have zero mean and unit variance.

- Let μ_j be the mean of feature j:

$$\mu_j = \frac{1}{n}\sum_{i=1}^{n} x_j^{(i)}$$

- Let s_j be the standard deviation of feature j:

$$\sigma_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_j^{(i)} - \mu_j\right)^2}$$

- Replace each feature j with:

$$x_j^{(i)} = \frac{x_j^{(i)} - \mu_j}{\sigma_j}$$

Question: How do we perform the transformation on training, validation and test set?

Feature Scaling – Mean normalization

Mean normalization rescales features to value range between $[-1,1]$.

- Let μ_j be the mean of feature j: $\mu_j = \frac{1}{n}\sum_{i=1}^{n} x_j^{(i)}$.
- Let $\max_i x_j$ be the maximum value of feature j of all data samples.
- Let $\min_i x_j$ be the minimum value of feature j of all samples.
- Replace each feature j with:

$$x_j^{(i)} = \frac{x_j^{(i)} - \mu_j}{\max_i x_j - \min_i x_j}$$

- This distribution will have the value range between $[-1,1]$.

Example: $x^{(1)} = [1,2]$ and $x^{(2)} = [3,4]$; We have $\mu_1 = 2, \mu_2 = 3, \max_i x_1 = 3,$ $\min_i x_1 = 1, \max_i x_2 = 4, \min_i x_2 = 2$

$x_1^{(1)} = \frac{1-2}{3-1} = -\frac{1}{2}; x_2^{(1)} = \frac{2-3}{4-2} = -\frac{1}{2}; x_1^{(2)} = \frac{3-2}{3-1} = \frac{1}{2}; x_2^{(2)} = \frac{4-3}{4-2} = \frac{1}{2}$

Therefore after mean normaliaztion, $x^{(1)} = \left[-\frac{1}{2}, -\frac{1}{2}\right]$ and $x^{(2)} = \left[\frac{1}{2}, \frac{1}{2}\right]$

Feature Scaling – Min-max scaling

Min-max scaling rescales features to the value range between $[0,1]$.

- Let $\max_i x_j$ be the maximum value of feature j of all data samples.
- Let $\min_i x_j$ be the minimum value of feature j of all data samples.
- Replace each feature j with:

$$x_j^{(i)} = \frac{x_j^{(i)} - \min_i x_j}{\max_i x_j - \min_i x_j}$$

Example: $x^{(1)} = [1,2]$ and $x^{(2)} = [3,4]$;

We have $\max_i x_1 = 3$, $\min_i x_1 = 1$, $\max_i x_2 = 4$, $\min_i x_2 = 2$

$x_1^{(1)} = \frac{1-1}{3-1} = 0$; $x_2^{(1)} = \frac{2-2}{4-2} = 0$; $x_1^{(2)} = \frac{3-1}{3-1} = 1$; $x_2^{(2)} = \frac{4-2}{4-2} = 1$

Therefore after Min-max scaling, $x^{(1)} = [0,0]$ and $x^{(2)} = [1,1]$

Unit vector scaling

Unit vector scaling rescales each feature so that the entire feature vector has a length of 1 (unit vector).

- Let $\left\|x^{(i)}\right\| = \sqrt{\sum_{j=1}^{n}\left(x_j^{(i)}\right)^2}$ be the length of the entire feature vector $x^{(i)}$.
- Replace each feature j with:

$$x_j^{(i)} = \frac{x_j^{(i)}}{\|x^{(i)}\|}$$

Example: $x^{(1)} = [1,2]$ and $x^{(2)} = [3,4]$;

We have $\left\|x^{(1)}\right\| = \sqrt{5}$, $\left\|x^{(2)}\right\| = 5$

$x_1^{(1)} = \frac{1}{\sqrt{5}}$; $x_2^{(1)} = \frac{2}{\sqrt{5}}$; $x_1^{(2)} = \frac{3}{5}$; $x_2^{(2)} = \frac{4}{5}$. Therefore after Unit vector scaling, $x^{(1)} = \left[\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right]$, $x^{(2)} = \left[\frac{3}{5}, \frac{4}{5}\right]$, such that $\left\|x^{(1)}\right\| = \left\|x^{(2)}\right\| = 1$

Practice

Mean normalization

$$x_j^{(i)} = \frac{x_j^{(i)} - \mu_j}{\max_i x_j - \min_i x_j}$$

Min-max scaling

$$x_j^{(i)} = \frac{x_j^{(i)} - \min_i x_j}{\max_i x_j - \min_i x_j}$$

Unit vector scaling

$$x_j^{(i)} = \frac{x_j^{(i)}}{\|x^{(i)}\|}$$

Practice question: there are three data samples

$$x^{(1)} = [1, 2, 3];$$
$$x^{(2)} = [4, 5, 6];$$
$$x^{(3)} = [9, 8, 7];$$

Scale these data samples using three methods: Mean normalization, Min-max scaling and Unit vector scaling.

Regularization

Question: With the same dataset, you trained and obtained two models as follows:

- Model 1: $h_\theta(x) = 1078 + 19254x - 3115x^2 - 982x^3$
- Model 2: $h_\theta(x) = 4 - 12x - 7x^2 + 13x^3$

Which model do you believe is more likely to exhibit better generalization to new data samples?

Regularization

Regularization is a technique used in machine learning and statistics to prevent overfitting and improve the generalization performance of a predictive model.

- *L1 Regularization (Lasso regression)*: L1 regularization adds a penalty term proportional to the *absolute value* of the model's coefficients (θs) to the objective function.
- *L2 Regularization (Ridge regression)*: L2 regularization adds a penalty term proportional to the *squared value* of the model's coefficients (θs) to the objective function.

In Ridge regression, L2 regularization imposes a stronger penalty (squared) on the magnitude of the model's coefficients.

Regularization – Objective function

Objective function of linear regression with L2 regularization:

$$J(\theta) = \underbrace{\frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2}_{\text{Model fit to data}} + \underbrace{\frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2}_{\text{L2 regularization}}$$

- λ is the regularization parameter ($\lambda \geq 0$)
- No regularization on bias θ_0

Questions:

- What happens if λ is large?
- What happens if λ is small?

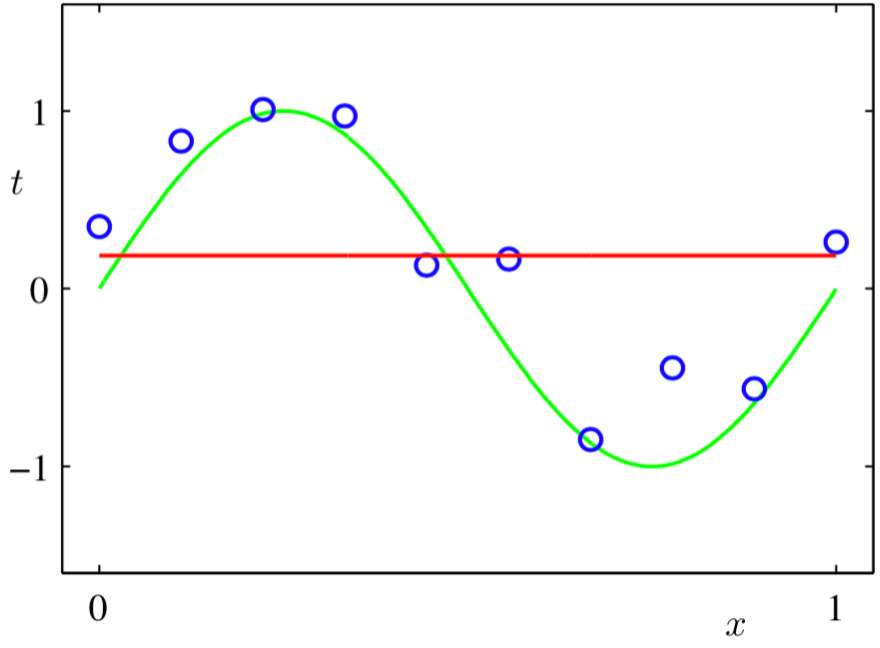
Regularization – Objective function

Objective function of linear regression with L2 regularization:

$$J(\theta) = \underbrace{\frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2}_{\text{Model fit to data}} + \underbrace{\frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2}_{\text{L2 regularization}}$$

Questions:

- What happens if λ is large?
  - θ_j tends to be small
  - $h_\theta(x) \approx \theta_0$
  - The model is …



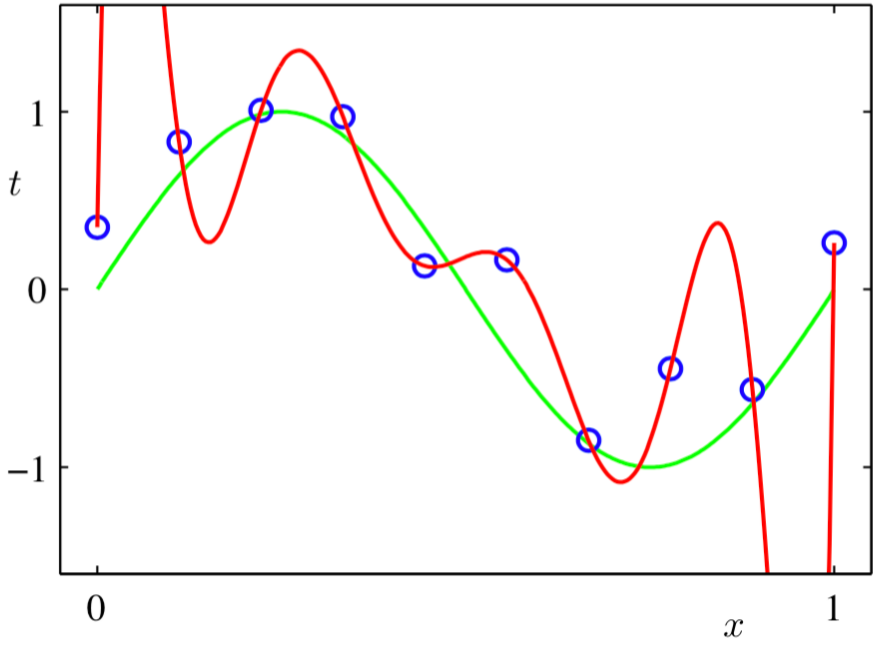
Regularization – Objective function

Objective function of linear regression with L2 regularization:

$$J(\theta) = \underbrace{\frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2}_{\text{Model fit to data}} + \underbrace{\frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2}_{\text{L2 regularization}}$$

Questions:

- What happens if λ is small?
  - θ_j tends to be large
  - The model is …



Regularization – Parameter update

Objective function of linear regression with L2 regularization:

$$J(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2$$

Fitting by solving:

$$\min_\theta J(\theta)$$

Parameter update:

$$\theta_j = \theta_j - \alpha \underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_j^{(i)} + \lambda\theta_j\right)}_{\text{Gradient } \frac{\partial}{\partial\theta_j}J(\theta)}$$

Summary

- Linear regression
- Loss Function
- Gradient descent
    - Batch Gradient Descent
    - Stochastic Gradient Descent
    - Mini-batch Gradient Descent
- Learning Rate
- Underfitting & Overfitting
- Generalization using Training, Validation and Test set
- Feature scaling
    - Standardization
    - Mean Normalization
    - Min-Max scaling
    - Unit Vector
- Regularization

Q&A

Thank you